Shared–idiosyncratic frailty models for multivariate count data

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ABSTRACT

Multivariate count data frequently exhibit both cross-sectional dependence and marginal overdispersion, posing challenges for statistical modeling and interpretation. While copula-based constructions and latent Gaussian models provide flexible dependence structures, they often involve computational complexity or lack a clear separation between sources of dependence and marginal heterogeneity. We propose a computationally tractable class of multivariate count models based on a shared–idiosyncratic frailty decomposition, in which dependence and marginal overdispersion are generated by distinct latent components. Conditional on latent variables, responses follow independent Poisson distributions whose intensities factorize into a common shared frailty and component-specific effects. This formulation yields an identifiable latent structure with interpretable parameters while preserving mixed-Poisson marginal behavior. We derive distributional properties through the joint probability generating function and develop likelihood-based inference using an efficient EM algorithm whose computational cost scales linearly with sample size and dimension. Extensive simulation studies compare the proposed approach with shared-frailty, Poisson–lognormal, and copula-based models, demonstrating improved recovery of dependence features beyond second-order moments and competitive likelihood performance under both correct specification and model misspecification. An application to epileptic seizure count data illustrates how the decomposition provides interpretable summaries of global and component-specific heterogeneity. The proposed framework offers a practical and interpretable alternative for modeling multivariate count data, combining structural interpretability with scalable computation.

1. Introduction

Existing multivariate count models generally struggle to separate marginal heterogeneity from cross-sectional dependence in an identifiable manner. In many commonly used constructions, overdispersion and dependence arise from the same latent mechanism, making their respective contributions difficult to interpret and, in some cases, statistically non-identifiable Hougaard (2000); Molenberghs and Verbeke (2005); Aalen, Borgan and Gjessing (2008). Similar issues arise in copula-based models for discrete data, where non-uniqueness and interpretation of dependence parameters may complicate structural inference Genest and Nešlehová (2007); Panagiotelis, Joe and Czado (2012b). This limitation motivates the search for modeling frameworks in which these two sources of variability admit a principled structural decomposition.

A fundamental statistical question underlying these difficulties is whether cross-sectional dependence and marginal overdispersion can in fact be separated in an identifiable manner within multivariate mixed Poisson models. While many constructions introduce multiple latent components, it is generally unclear whether the resulting decomposition represents a genuine statistical structure or merely a reparameterization producing observationally equivalent distributions. Establishing identifiability of such a decomposition is therefore essential for meaningful interpretation of latent mechanisms.

Multivariate count data arise in a wide range of applied fields, including biomedical research, economics and finance, insurance, reliability analysis, and environmental monitoring. Representative examples include recurrent event counts recorded across multiple anatomical sites, transaction counts for groups of related assets, and numbers of failures observed on interconnected system components. Such data have long motivated the development of multivariate

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discrete distributions and counting-process models, ranging from early constructions of the bivariate Poisson distribution Holgate (1964) and classical treatments of multivariate discrete laws Kocherlakota and Kocherlakota (1992) to modern applications in sports analytics, recurrent event analysis, and event-history modeling Karlis and Ntzoufras (2003); Cook and Lawless (2007); Cox (1972); Aalen et al. (2008). In these settings, statistical modeling requires the simultaneous accommodation of cross-sectional dependence, marginal overdispersion, and parameter identifiability, while preserving computational feasibility as the dimension increases.

A widely adopted strategy combines Poisson or negative binomial marginal models with copula-based dependence structures. Copula constructions have been extensively proposed for modeling multivariate discrete responses and count data Nelsen (2006); Joe (2014); Joe, Li and Nikoloulopoulos (2010); Nikoloulopoulos and Karlis (2009); Nikoloulopoulos (2013). Although copulas provide a general and flexible framework for modeling dependence, their application to discrete data is often accompanied by practical and theoretical challenges, including non-uniqueness, potential identifiability issues, and rapidly increasing computational complexity in higher dimensions Panagiotelis et al. (2012b). Moreover, copula parameters typically summarize dependence in an abstract manner, without a direct generative interpretation, which may limit their usefulness in applications where latent mechanisms are of primary scientific interest.

Latent-variable formulations offer an alternative approach by inducing dependence through shared random effects or frailty components. Mixed Poisson and frailty models have long been used to account for unobserved heterogeneity and overdispersion, and numerous multivariate extensions have been developed Becker and Utev (2002); Peyhardi, Fernique and Durand (2021); Fokianos, Fried, Kharin and Voloshko (2022). While these models provide appealing generative interpretations, many existing specifications confound marginal heterogeneity with cross-sectional dependence or encounter identifiability difficulties when multiple latent scales are introduced. To our knowledge, no identifiable decomposition simultaneously separating dependence and marginal dispersion within a unified multivariate count framework has been formally established.

The primary objective of this paper is therefore theoretical. Our main contribution is an identifiability theorem showing that cross-sectional dependence and marginal overdispersion can be uniquely separated within a broad class of multivariate mixed Poisson models. Under mild normalization conditions and independence assumptions between latent components, we prove that the joint distribution uniquely determines a decomposition into a shared factor generating cross-sectional dependence and component-specific factors governing marginal overdispersion.

In this paper, we propose a multivariate count model based on a shared–idiosyncratic frailty decomposition, in which dependence and marginal heterogeneity are generated by distinct latent components. Conditionally on latent variables, each response follows a Poisson distribution whose intensity factorizes into a common multiplicative frailty, inducing cross-sectional dependence, and a component-specific frailty governing marginal overdispersion. Given these latent variables, the observed counts are independent.

The proposed construction should thus be viewed not merely as a new model specification, but as a constructive probabilistic representation achieving this identifiable decomposition. When the idiosyncratic frailty components are specified within the Gamma family, marginalization over the component-specific effects yields negative binomial margins with a shared frailty. Consequently, the proposed framework does not introduce a new marginal count distribution. Instead, its primary contribution lies in providing an identifiable and interpretable latent decomposition that separates the mechanisms responsible for cross-sectional dependence from those governing marginal heterogeneity.

This perspective enables a transparent analysis of shared and idiosyncratic sources of variability in multivariate count data and facilitates a detailed characterization of the dependence structure induced by a common latent multiplier. In particular, the proposed model admits tractable expressions for joint moments and dependence measures through its probability generating function and supports formal identifiability under mild normalization constraints. The explicit latent structure further enables likelihood-based inference via an EM algorithm and promotes clear interpretation of model parameters.

Accordingly, the contribution of this work is primarily theoretical rather than distributional: we establish a general identifiability theorem for a latent decomposition separating dependence and dispersion, while the shared–idiosyncratic frailty formulation provides an explicit and tractable realization of this structure.

The main contributions of this paper are threefold. First, we establish an identifiability result showing that cross-sectional dependence and marginal overdispersion admit a separable and identifiable decomposition within a broad class of multivariate mixed Poisson models under mild normalization conditions. Second, we introduce a shared–idiosyncratic frailty construction that provides an explicit probabilistic representation realizing this identifiable structure while retaining analytical tractability. Third, we develop likelihood-based inference procedures based on

an EM algorithm and demonstrate, through simulation studies and a real data application, how the identifiable decomposition improves recovery and interpretation of dependence features beyond second-order moments.

The remainder of the paper is organized as follows. Section 2 introduces the proposed model and discusses its relation to existing multivariate count constructions. Section 3 develops distributional properties and characterizes the induced dependence structure. Identifiability and inference are addressed in Sections 3.6 and 3.8, respectively. Section 4 reports simulation studies investigating finite-sample estimation, recovery of dependence features beyond second-order moments, and robustness of the proposed model relative to alternative multivariate count formulations. Section 5 illustrates the methodology through an empirical application to epileptic seizure count data, and Section 6 concludes the paper. Technical proofs are deferred to the Appendix.

2. Model construction and special cases

Let $\mathbf{Y} = (Y_1, \dots, Y_d)^\top$ be a d-dimensional count vector taking values in $\mathbb{N}_0^d$. Throughout the paper, $\mathbb{N}_0 := \{0, 1, 2, \dots\}$ denotes the set of nonnegative integers, and $\mathbb{N}_0^d$ its d-fold Cartesian product. We develop a flexible family of multivariate count models based on a shared–idiosyncratic frailty decomposition that separates global dependence from component-specific heterogeneity in an interpretable way. Shared frailty constructions of this type originate in survival and event-history analysis, where multiplicative latent effects induce dependence between outcomes (Clayton, 1978; Duchateau and Janssen, 2008).

Let $U > 0$ denote a common (shared) frailty and let $V_1, \dots, V_d > 0$ denote idiosyncratic frailty terms. Assume that

$$U \sim G_0, \qquad V_j \sim G_j, \quad j = 1, \dots, d,$$

where $U, V_1, \dots, V_d$ are mutually independent and $G_0, G_1, \dots, G_d$ are probability measures supported on $(0, \infty)$.

When parametric frailty distributions are employed, we focus throughout on a Gamma specification. Specifically, we adopt the shape–rate parameterization: a random variable $X \sim \text{Gamma}(a, b)$ has density

$$f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}, \qquad x > 0,$$

so that $\mathbb{E}(X) = a/b$ and $\text{Var}(X) = a/b^2$. Gamma frailty distributions are widely used due to their analytical tractability and their ability to yield closed-form mixture representations in multiplicative random-effect models (Duchateau and Janssen, 2008; Karlis and Xekalaki, 2005). Unless otherwise stated, the shared frailty and the idiosyncratic frailties are assumed to follow

$$U \sim \text{Gamma}(\alpha, \alpha), \qquad V_j \sim \text{Gamma}(\beta_j, \beta_j), \quad j = 1, \dots, d,$$

so that $\mathbb{E}(U) = \mathbb{E}(V_j) = 1$. We refer to this specification as the Gamma–Gamma shared–idiosyncratic frailty model. The unit-mean normalization ensures identifiability of the baseline intensity parameters λ_j and facilitates interpretation of dependence and dispersion effects. Conditional on $(U, \mathbf{V}) = (u, v_1, \dots, v_d)$, the coordinates of $\mathbf{Y}$ are independent and satisfy

$$Y_j \mid (U = u, V_j = v_j) \sim \text{Poisson}(\lambda_j u v_j), \qquad j = 1, \dots, d, \tag{1}$$

where $\lambda_j > 0$ are baseline intensity parameters.

Under this specification, cross-component dependence is induced exclusively through the shared frailty U, whereas the V_j's govern marginal heterogeneity and overdispersion in each coordinate. The distinction is deliberate: U represents system-wide shocks that act on all components simultaneously, while V_j captures residual variation specific to component j. Compared with conventional shared-random-effect or factor-type count models, the proposed decomposition assigns separate latent factors to dependence and marginal dispersion, which clarifies interpretation and supports identifiable inference. In what follows, we establish that this construction yields a broad class of multivariate count distributions with tractable moments and dependence measures, and it accommodates flexible choices of G_0 and $\{G_j\}_{j=1}^d$ suited to particular applications.

2.1. Joint distribution and probability generating function

By iterated conditioning, the joint probability mass function of $\mathbf{Y}$ can be written as the nested mixture

$$\mathbb{P}(\mathbf{Y} = \mathbf{y}) = \int_0^\infty \prod_{j=1}^d \left\{ \int_0^\infty \frac{\exp(-\lambda_j u v)\,(\lambda_j u v)^{y_j}}{y_j!}\, dG_j(v) \right\} dG_0(u), \qquad \mathbf{y} \in \mathbb{N}_0^d. \tag{2}$$

Representations of this form belong to the general class of mixed Poisson distributions, which provide a flexible mechanism for introducing overdispersion and dependence through latent mixing variables (Karlis and Xekalaki, 2005). Let

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\prod_{j=1}^d s_j^{Y_j}\right], \qquad \mathbf{s} \in [0,1]^d,$$

denote the multivariate probability generating function (pgf). Using conditional independence given $(U, \mathbf{V})$ and the pgf of a Poisson distribution, we obtain

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\exp\left\{ U \sum_{j=1}^d \lambda_j V_j (s_j - 1) \right\}\right]. \tag{3}$$

Representation (3) is a key analytical device: it yields joint moments and dependence measures by differentiation and supports identifiability arguments via uniqueness of Laplace transforms. Moreover, (3) clarifies how the proposed construction spans a broad class of multivariate count distributions under flexible choices of G_0 and $\{G_j\}_{j=1}^d$. In the next subsection, we demonstrate that several familiar multivariate count models are recovered as special cases of this shared–idiosyncratic frailty framework.

2.2. Special cases and connections to existing models

The shared–idiosyncratic frailty construction offers a convenient organizing principle for multivariate count models: the common factor U governs cross-component association, while $\{V_j\}_{j=1}^d$ controls marginal heterogeneity. Several familiar specifications are obtained by restricting these latent components, and the framework clarifies how such restrictions limit the dependence patterns that can be expressed.

A first limiting case corresponds to a degenerate shared frailty, $U \equiv 1$. In this situation, the coordinates $\{Y_j\}$ are independent mixed Poisson variables, with mixing distributions induced solely by the idiosyncratic frailties $\{V_j\}$. All overdispersion is therefore marginal, and no cross-component dependence is present.

At the opposite extreme, when the idiosyncratic frailties are degenerate, $V_j \equiv 1$ for all j, the model reduces to

$$Y_j \mid U = u \sim \text{Poisson}(\lambda_j u), \qquad j = 1, \dots, d,$$

so dependence is generated entirely through the common multiplier U. Appropriate choices of the distribution G_0 connect this specification to classical multivariate Poisson constructions based on shared components or reduction devices (see, e.g., Holgate, 1964; Karlis, 2003). . Unlike purely discrete shared-shock representations, allowing U to be continuously distributed yields a broader range of dependence strengths and tail behaviors while preserving a transparent generative interpretation.

More generally, when $\log U$ and $\log V_j$ are Gaussian (or admit Gaussian factor structures), the proposed model specializes to Poisson–lognormal-type mixtures and closely related factor-analytic count models (Aitchison and Ho, 1989; Subedi and Dang, 2025). The present formulation keeps the roles of U and V_j distinct, which facilitates interpretation and, as shown later, leads to tractable expressions for moments and dependence measures under mild regularity conditions.

Finally, it is worth contrasting the proposed construction with copula-based approaches to multivariate count modeling (Nelsen, 2006). While copulas offer great flexibility in the continuous setting, discreteness complicates identifiability and may lead to non-unique dependence representations (Genest and Nešlehová, 2007). In the present framework, dependence is induced through an explicit data-generating mechanism, so parameters map directly to interpretable features: λ_j controls baseline intensity, G_0 governs cross-sectional association, and G_j determines marginal dispersion. This separation also supports scalable modeling in higher dimensions.

Taken together, these special cases motivate the proposed framework as a flexible extension of widely used multivariate count models, while providing a clear baseline for interpreting how assumptions on G_0 and $\{G_j\}$ shape both marginal behavior and dependence.

2.3. Relation to shared-frailty Negative Binomial models

Consider the shared–idiosyncratic frailty model introduced in Section 2, in which, conditional on latent variables $(U, V_1, \dots, V_d)$, the components Y_j are independent and follow

$$Y_j \mid (U = u, V_j = v_j) \sim \text{Poisson}(\lambda_j u v_j), \quad j = 1, \dots, d,$$

with a common frailty U and component-specific frailties V_j. When the idiosyncratic frailties V_j are assumed to follow Gamma distributions with unit mean, marginalizing out V_j yields

$$Y_j \mid U = u \sim \text{Negative Binomial}(\lambda_j u, \beta_j),$$

where β_j governs marginal overdispersion. Consequently, marginalizing over both the shared and idiosyncratic frailties leads to a multivariate count model belonging to the class of shared-frailty negative binomial models.

This observation implies that the proposed framework does not introduce a new marginal count distribution when Gamma idiosyncratic frailties are adopted. Rather, it provides a latent-variable representation of a shared-frailty negative binomial model in which marginal overdispersion and cross-sectional dependence arise from distinct latent components. From this perspective, the shared–idiosyncratic formulation can be viewed as a structured generative decomposition of a classical model, rather than as an extension of its marginal family.

The distinction is nevertheless important. In standard shared-frailty negative binomial models, overdispersion and dependence are both absorbed into the shared random effect, potentially confounding component-specific heterogeneity with cross-sectional association. By contrast, the present construction separates these sources of variability explicitly. This separation yields parameters with a clear probabilistic interpretation and supports identifiability of the latent structure under mild normalization constraints. In particular, the probability generating function factorization developed in Section 3 allows the shared and idiosyncratic components to be identified from the joint distribution, a property that is not immediate in more aggregated representations.

2.4. Relation to Poisson–lognormal models

Poisson–lognormal models constitute a popular and flexible class of latent-variable models for multivariate count data. In these formulations, dependence and overdispersion are induced through a Gaussian latent vector, whose covariance structure controls both marginal variability and cross-sectional association. By suitably specifying this covariance matrix, Poisson–lognormal models can accommodate a wide range of dependence patterns, including shared and component-specific effects. More generally, when $\log U$ and $\log V_j$ are Gaussian (or admit Gaussian factor structures), the proposed model specializes to Poisson–lognormal-type mixtures and closely related factor-analytic count models (Aitchison and Ho, 1989; Subedi and Dang, 2025).

The shared–idiosyncratic frailty framework considered here differs primarily in its modeling philosophy. Rather than relying on a flexible covariance structure in a Gaussian latent space, it introduces distinct latent variables with explicit multiplicative effects on the Poisson intensities. The shared frailty governs cross-sectional dependence, while the idiosyncratic frailties control marginal overdispersion. This separation leads to a parsimonious parameterization with a direct generative interpretation, in which dependence is induced exclusively through variability in the shared latent multiplier.

From a methodological standpoint, neither approach dominates the other in general. Poisson–lognormal models may achieve greater flexibility in representing heterogeneous dependence structures, especially when the covariance matrix of the latent Gaussian vector is richly parameterized. However, this flexibility may come at the cost of interpretability and identifiability of shared versus component-specific effects. In contrast, the proposed frailty-based decomposition imposes a more structured form of dependence, but yields parameters that map directly to interpretable latent mechanisms and supports tractable likelihood-based inference.

2.5. Implications for modeling and inference

The above discussion highlights that the contribution of the proposed shared–idiosyncratic frailty model is not distributional novelty, but conceptual clarity. By making explicit the roles of shared and idiosyncratic latent components, the framework facilitates interpretation of dependence and overdispersion in multivariate count data and provides a transparent basis for inference. The model is particularly well suited to applications in which a common latent factor is believed to drive simultaneous variation across components, while allowing for substantial heterogeneity at the marginal level.

At the same time, the structured nature of the dependence implies limitations. Because dependence is induced solely through the shared frailty, the model primarily captures positive association of a common-factor type and may be less effective in settings requiring highly heterogeneous or component-specific dependence patterns. In such cases, Poisson–lognormal or copula-based approaches may offer greater flexibility. The proposed framework should therefore be viewed as a complementary tool within the broader landscape of multivariate count models, rather than as a universal solution. Latent-variable formulations of this kind provide an explicit generative interpretation of dependence, in contrast with purely distributional constructions (Clayton, 1978).

3. Distributional properties and dependence structure

This section investigates the probabilistic implications of the shared–idiosyncratic frailty decomposition introduced in Section 2. While several properties derived below resemble classical results for mixed Poisson models, their interpretation here is fundamentally structural: the separation between shared and idiosyncratic frailties allows each aspect of dispersion and dependence to be attributed to a distinct latent mechanism. The results therefore characterize not merely a mixture representation, but the probabilistic consequences of an identifiable latent decomposition. The focus is on clarifying how marginal overdispersion and cross-sectional dependence arise from distinct latent components, and on characterizing the dependence structure induced by the shared frailty. Throughout, the results should be interpreted in light of the positioning discussed in Section 2.3.

3.1. Marginal distributions

We first consider the one-dimensional behavior of each component. Although cross-sectional dependence is induced through the shared frailty U, each marginal distribution remains within the class of mixed Poisson models, with mixing driven jointly by the shared and idiosyncratic latent variables.

Proposition 1. *For each $j = 1, \dots, d$, the marginal distribution of Y_j is mixed Poisson with random mean $\lambda_j U V_j$, that is,* $Y_j \,\Big|\, (U, V_j) \sim \text{Poisson}(\lambda_j U V_j) \quad \Longrightarrow \quad Y_j \sim \text{MixPois}(\lambda_j U V_j)$.

The proof of Proposition 1 and all other proofs can be found in the Appendix A

Here and throughout, MixPois(Λ) denotes a mixed Poisson distribution, namely a Poisson distribution conditionally on a nonnegative random intensity Λ, with the unconditional law obtained by mixing over the distribution of Λ. Equivalently, $Y \sim \text{MixPois}(\Lambda)$ if

$$\mathbb{P}(Y = y) = \mathbb{E}\left[e^{-\Lambda}\frac{\Lambda^y}{y!}\right], \qquad y \in \mathbb{N}_0.$$

An immediate implication is that $\mathbb{E}(Y_j) = \lambda_j\,\mathbb{E}(U)\,\mathbb{E}(V_j)$, and the variance admits the decomposition $\text{Var}(Y_j) = \mathbb{E}(Y_j) + \lambda_j^2\,\text{Var}(UV_j)$, where, under independence of U and V_j,

$$\text{Var}(UV_j) = \text{Var}(U)\,\mathbb{E}(V_j)^2 + \mathbb{E}(U)^2\,\text{Var}(V_j) + \text{Var}(U)\,\text{Var}(V_j).$$

These expressions highlight the distinct roles of the latent components: variability in the shared frailty inflates dispersion uniformly across margins, whereas idiosyncratic frailties contribute additional component-specific overdispersion. This separation mirrors the latent decomposition discussed in Section 2.3. In contrast, in standard mixed Poisson constructions driven by a single latent variable, marginal dispersion and cross-sectional dependence cannot generally be disentangled. Here, the variance decomposition explicitly separates a global contribution driven by U from component-specific variability induced by V_j, a distinction that will play a central role in the identifiability results developed later.

3.2. Second-order moments and covariance structure

We now examine second-order dependence properties. Because the components of $\mathbf{Y}$ are conditionally independent given $(U, \mathbf{V})$, cross-sectional dependence arises solely through variation in the shared frailty. This yields a transparent expression for pairwise covariance via the law of total covariance.

Theorem 1. *For any $j \neq k$,*

$$\text{Cov}(Y_j, Y_k) = \lambda_j \lambda_k\,\text{Var}(U)\,\mathbb{E}(V_j)\,\mathbb{E}(V_k).$$

Theorem 1 shows that the common factor U is the sole source of contemporaneous dependence: the covariance vanishes if and only if $\mathrm{Var}(U) = 0$, whereas the idiosyncratic frailties affect the marginal scales through $\mathbb{E}(V_j)$ but do not generate dependence across distinct components. Moreover, since $\lambda_j, \lambda_k > 0$ and $\mathbb{E}(V_j), \mathbb{E}(V_k) > 0$, it follows that $\mathrm{Cov}(Y_j, Y_k) \geq 0$, with strict inequality whenever $\mathrm{Var}(U) > 0$. The proof of Theorem 1 is provided in Appendix A.

Proposition 2. *Under the normalization* $\mathbb{E}(U) = \mathbb{E}(V_j) = 1$*, the pairwise covariance simplifies to* $\mathrm{Cov}(Y_j, Y_k) = \lambda_j \lambda_k \mathrm{Var}(U)$*. Consequently, the correlation coefficient satisfies*

$$\mathrm{Corr}(Y_j, Y_k) = \frac{\lambda_j \lambda_k \mathrm{Var}(U)}{\sqrt{\mathrm{Var}(Y_j)\mathrm{Var}(Y_k)}}.$$

In particular, cross-sectional dependence vanishes if and only if $\mathrm{Var}(U) = 0$.

Proposition 2 highlights a key structural feature of the proposed model: the strength of cross-sectional dependence is entirely governed by the variability of the shared frailty U. In particular, $\mathrm{Var}(U)$ acts as a direct driver of dependence between the count variables, whereas the idiosyncratic frailties influence only marginal dispersion. The proof of Proposition 2 is provided in Appendix A.

As discussed in Section 2.3, this structure contrasts with Poisson–lognormal models, where covariance typically reflects both shared and component-specific effects through the covariance matrix of a Gaussian latent vector. In such formulations, several latent components may contribute simultaneously to the covariance structure, making dependence parameters more difficult to interpret.

By contrast, the shared–idiosyncratic decomposition isolates $\mathrm{Var}(U)$ as the unique driver of cross-sectional association, while the distributions of the idiosyncratic frailties govern marginal overdispersion. This separation provides a transparent interpretation of dependence and dispersion and forms the basis for the identifiability results developed in Section 3.6.

3.3. Multivariate probability generating function

A key analytical tool underlying the results of this paper is the multivariate probability generating function (pgf) of $\mathbf{Y}$. Beyond providing access to joint moments, the pgf plays a central role in characterizing dependence and establishing identifiability of the latent decomposition.

Proposition 3. *The multivariate probability generating function of* $\mathbf{Y}$ *is given by*

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\exp\left\{U \sum_{j=1}^{d} \lambda_j V_j (s_j - 1)\right\}\right], \qquad \mathbf{s} \in [0,1]^d.$$

Proposition 3 shows that the joint pgf can be expressed as an expectation over the latent frailty variables. This representation reveals the multiplicative interaction between the shared frailty U and the idiosyncratic frailties V_j, which together determine the joint distribution of the counts.

Proposition 4. *Let* $\mathbf{Y} = (Y_1, \dots, Y_d)^\top$ *follow the shared–idiosyncratic frailty model* $Y_j \mid (U, V_j) \sim \mathrm{Poisson}(\lambda_j U V_j), \quad j = 1, \dots, d$*, where* $U, V_1, \dots, V_d$ *are mutually independent. Let* $L_j(t) = \mathbb{E}\left[e^{-tV_j}\right], \quad t \geq 0$*, denote the Laplace transform of* V_j*, and let* $L_U(t) = \mathbb{E}\left[e^{-tU}\right], \quad t \geq 0$*, denote the Laplace transform of* U*. Then the joint probability generating function of* $\mathbf{Y}$ *admits the representation*

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\prod_{j=1}^{d} L_j\big(\lambda_j U (1 - s_j)\big)\right], \quad \mathbf{s} \in [0,1]^d.$$

Equivalently,

$$P_{\mathbf{Y}}(\mathbf{s}) = \int_0^\infty \prod_{j=1}^{d} L_j\big(\lambda_j u (1 - s_j)\big)\, dG_0(u).$$

In particular, the shared frailty distribution G_0 *enters only through the outer mixing over* u*, whereas the idiosyncratic frailty distributions* $\{G_j\}_{j=1}^d$ *enter through component-specific Laplace transforms. Hence the joint pgf separates the common source of dependence from the marginal sources of overdispersion.*

Proposition 4 shows that the proposed model is characterized by a nested Laplace-transform structure in which the shared frailty acts as a global mixing variable, while the idiosyncratic frailties act componentwise through marginal Laplace transforms. This factorization is not a generic feature of multivariate mixed Poisson models. It is precisely this structural separation that makes it possible to distinguish cross-sectional dependence from marginal overdispersion and underlies the identifiability result established in Theorem 4. The proofs of Propositions 3 and 4 are provided in Appendix A.

3.4. Positive dependence and association

Beyond covariance, it is important to characterize qualitative dependence properties implied by the model. In particular, we show that the shared–idiosyncratic frailty construction induces positive dependence in the sense of association.

Theorem 2 (Association). *The random vector $Y = (Y_1, \dots, Y_d)^\top$ generated by the shared–idiosyncratic frailty model is associated in the sense of (Esary, Proschan and Walkup, 1967). That is, for any coordinatewise non-decreasing functions f and g for which the covariance exists,*

$$\mathrm{Cov}(f(Y), g(Y)) \geq 0.$$

The proof relies on a closure property of association under monotone mixing with respect to a one-dimensional latent variable and is given in Appendix A.

Corollary 1 (Positive quadrant dependence). *The random vector Y is positively quadrant dependent. In particular, for any $j \neq k$ and all $(x, y) \in \mathbb{R}^2$,*

$$\mathbb{P}(Y_j > x,\, Y_k > y) \;\geq\; \mathbb{P}(Y_j > x)\,\mathbb{P}(Y_k > y).$$

PROOF. Association implies positive quadrant dependence; see, for example, (Shaked and Shanthikumar, 2007, Section 1.7).

Such forms of positive dependence are not generically captured by models relying solely on linear correlation and typically require additional structural assumptions when Gaussian latent variables are employed. The association property follows directly from the presence of a single monotone shared multiplier. Unlike constructions in which dependence is introduced through several latent factors, the sign and qualitative nature of dependence are structurally determined by the shared frailty alone.

3.5. Beyond second-order dependence

While second-order moments provide useful summaries, they are often insufficient for describing dependence in multivariate count data with latent heterogeneity. We therefore briefly consider rank-based and tail dependence measures to further elucidate the dependence structure induced by the shared frailty.

We first consider Kendall's tau, a rank-based measure of dependence capturing monotone association beyond linear correlation. For two components Y_j and Y_k, Kendall's tau is defined as

$$\tau_{jk} = \mathbb{P}\Big\{(Y_j - Y_j')(Y_k - Y_k') > 0\Big\} - \mathbb{P}\Big\{(Y_j - Y_j')(Y_k - Y_k') < 0\Big\},$$

where (Y_j', Y_k') denotes an independent copy of (Y_j, Y_k).

Under the proposed model, conditional on the shared frailty U, the components of $\mathbf{Y}$ are independent, so all dependence arises through mixing with respect to the common latent variable. Moreover,

$$\mathbb{E}[Y_j \mid U] = \lambda_j U \mathbb{E}[V_j],$$

which is strictly increasing in U. The dependence between Y_j and Y_k therefore arises through a common monotone transformation of the latent factor.

Proposition 5. *Under the shared–idiosyncratic frailty model defined in Section 2, the pairwise Kendall's tau between Y_j and Y_k satisfies $\tau_{jk} = 0$ if $\mathrm{Var}(U) = 0$, and $\tau_{jk} > 0$ whenever $\mathrm{Var}(U) > 0$. Hence, the shared frailty U is the unique source of pairwise concordance between the count variables.*

Proposition 5 formalizes the intuition that the strength of rank-based dependence is entirely driven by the variability of the shared frailty. When the shared frailty becomes degenerate, the components of $\mathbf{Y}$ become independent and Kendall's tau vanishes. Conversely, any non-degenerate shared frailty induces positive concordance across components. General results on concordance for latent factor models can be found in (Joe, 1997, Chapter 2).

We next turn to tail dependence, which describes the tendency of components to exhibit joint extreme values. Tail dependence properties depend critically on the tail behavior of the shared frailty. Accordingly, we state the following result under a heavy-tailed assumption on U.

Theorem 3. *Assume that the shared frailty U has a regularly varying tail with index $\rho > 0$, that is,*

$$\mathbb{P}(U > u) = u^{-\rho} L(u), \quad u \to \infty,$$

where L is a slowly varying function. Assume further that $\mathbb{E}[V_j] < \infty$ and $\mathbb{E}[V_k] < \infty$. Then the bivariate count vector (Y_j, Y_k) exhibits positive upper tail dependence, that is, $\lambda_U > 0$.

Proofs of Proposition 5 and Theorem 3, based on regular variation arguments for Poisson mixtures, are given in Appendix C.

Remark 1. *Positive upper tail dependence arises when the shared frailty has a heavy tail, due to its multiplicative effect on all components. In contrast, when the shared latent variable has light tails, such as in the commonly used Gamma–Gamma specification, the resulting Poisson mixtures are asymptotically tail independent. Nevertheless, Gamma frailties still induce substantial positive dependence at moderate levels through covariance, association, and rank-based measures. This distinction clarifies the scope of the tail dependence result and its relevance for practical modeling.*

In summary, the distributional and dependence properties derived in this section should be viewed as structural consequences of the shared–idiosyncratic decomposition rather than generic features of Poisson mixtures. Marginal overdispersion, second-order dependence, and stronger forms of positive association can all be traced back to interpretable features of the latent structure. These results complement the positioning in Section 2.3 and motivate the identifiability and inference developments that follow.

3.6. Main identifiability theorem

Identifiability is a fundamental requirement for meaningful statistical inference in latent-variable models, particularly in multivariate count settings where different latent structures may induce the same marginal likelihood. In the present framework, identifiability ensures that the proposed shared–idiosyncratic frailty decomposition represents a genuine statistical model rather than a mere reparameterization of existing formulations.

Identifiability in the proposed model relies on two key structural features. The first is the multiplicative decomposition of the Poisson intensities into shared and component-specific latent factors. The second is the factorization of the joint probability generating function induced by the assumed independence between the shared frailty and the idiosyncratic frailties. Together, these features allow the latent components to be uniquely recovered from the joint distribution of the observed counts, up to deterministic rescaling.

Theorem 4. *Let $\mathbf{Y} = (Y_1, \dots, Y_d)^\top$ follow the model defined in Section 2 with parameter $\Theta = (\lambda_1, \dots, \lambda_d, G_0, G_1, \dots, G_d)$, where $G_0, G_1, \dots, G_d$ are probability measures supported on $(0, \infty)$. Assume that $\mathbb{E}[U] = 1$ and $\mathbb{E}[V_j] = 1$ for all $j = 1, \dots, d$. Then the mapping $\Theta \longmapsto \mathcal{L}(\mathbf{Y})$ is identifiable.*

The proof exploits the uniqueness of multivariate Laplace transforms together with the multiplicative structure of the latent components. In particular, the joint probability generating function characterizes uniquely the distributions of the shared frailty and the idiosyncratic frailties under the imposed normalization constraints. A complete proof is given in Appendix B.

The following remarks clarify the scope, structural interpretation, and assumptions underlying the identifiability result.

Remark 2 (Scope of identifiability). *The identifiability result established in Theorem 4 is not restricted to the specific Gamma–Gamma specification considered later for inference. More generally, it applies to multiplicative mixed Poisson models whose joint probability generating function admits a factorization induced by independent shared*

and component-specific latent factors. The proposed shared–idiosyncratic frailty construction therefore provides a constructive representation of an identifiable subclass within multivariate mixed Poisson distributions, highlighting that identifiability arises from the structural factorization itself rather than from a particular parametric choice.

Remark 3 (Normalization and scale invariance). *The normalization conditions $\mathbb{E}[U] = 1$ and $\mathbb{E}[V_j] = 1$ are essential for identifiability and reflect an inherent scale invariance of the model. Without such constraints, the triples (λ_j, U, V_j) are identifiable only up to multiplicative reparameterizations, so that the joint distribution of $\mathbf{Y}$ determines an equivalence class of parameter values rather than a unique representative. The imposed normalization selects a convenient and interpretable parameterization within this equivalence class.*

Remark 4 (Extensions and alternative normalizations). *The identifiability argument based on the joint probability generating function readily extends to settings in which the baseline intensities λ_j depend on observed covariates through standard log-linear specifications, provided that the covariate design satisfies usual regularity conditions ensuring sufficient variation. Identifiability then follows conditionally on the covariates by the same Laplace transform arguments. Alternative normalization schemes are also possible, such as fixing $\mathbb{E}[U] = c$ and absorbing the scale change into the baseline intensities, or normalizing a single idiosyncratic frailty. These choices lead to equivalent identifiable parameterizations and do not affect the substance of Theorem 4, which holds modulo deterministic rescaling.*

Remark 5 (Role of independence assumptions). *Independence between the shared frailty U and the idiosyncratic frailties $\{V_j\}_{j=1}^d$ is a structural assumption underlying the identifiability result. Allowing dependence between these latent components would generally destroy the factorization properties of the joint Laplace transform and prevent separate recovery of shared and component-specific effects. In this sense, independence is not a technical convenience but a defining feature of the proposed decomposition.*

Taken together, Theorem 4 and the preceding remarks show that the shared–idiosyncratic frailty formulation defines an identifiable structural decomposition of latent variability rather than a model-specific parametrization. This identifiability provides a sound theoretical foundation for the likelihood-based inference procedures developed in the next section.

3.7. Likelihood formulation

This section outlines likelihood-based inference for the proposed shared–idiosyncratic frailty model. The estimation strategy follows standard principles for latent-variable models and is based on maximum likelihood. Although the inferential framework relies on a classical expectation–maximization (EM) approach, the presence of both shared and idiosyncratic frailties leads to a nonstandard complete-data structure. The purpose of this section is to clarify how the latent decomposition developed earlier supports tractable and interpretable inference, rather than to introduce a new estimation methodology.

Let $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ denote independent realizations of the d-dimensional count vector $\mathbf{Y}$. Under the model introduced in Section 2, the observed-data likelihood is obtained by integrating out the latent frailty variables.

For a generic observation $\mathbf{y}_i = (y_{i1}, \dots, y_{id})^\top$, the joint probability mass function is given by

$$p(\mathbf{y}_i; \Theta) = \int_0^\infty \prod_{j=1}^d \left\{ \int_0^\infty \frac{\exp(-\lambda_j u v)(\lambda_j u v)^{y_{ij}}}{y_{ij}!} \, dG_j(v) \right\} dG_0(u), \tag{4}$$

where $\Theta = (\lambda_1, \dots, \lambda_d, G_0, G_1, \dots, G_d)$ denotes the collection of model parameters.

The log-likelihood based on n independent observations is therefore

$$\ell_n(\Theta) = \sum_{i=1}^n \log p(\mathbf{y}_i; \Theta). \tag{5}$$

Direct maximization of (5) is generally infeasible due to the nested integrals over the latent variables. However, the explicit frailty representation naturally suggests an EM-type algorithm.

3.8. EM algorithm

Likelihood-based inference proceeds by treating the latent frailties as missing data. For each observation $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{id})$, introduce latent variables $(U_i, V_{i1}, \dots, V_{id})$, where U_i denotes the shared frailty and V_{ij} the idiosyncratic frailty associated with component j.

Conditional on these latent variables, the complete-data likelihood factorizes as

$$L_c(\Theta) = \prod_{i=1}^{n} \left[\prod_{j=1}^{d} \frac{\exp(-\lambda_j U_i V_{ij})(\lambda_j U_i V_{ij})^{y_{ij}}}{y_{ij}!} \right] dG_0(U_i) \prod_{j=1}^{d} dG_j(V_{ij}). \tag{6}$$

Inference is carried out via a generalized EM (GEM) algorithm, reflecting the joint presence of a shared frailty and multiple component-specific frailties. In the Gamma–Gamma specification considered throughout the paper, the latent variables U_i and V_{ij} are assumed independent and Gamma distributed with unit mean, as described in Section 2. At iteration t, the E-step evaluates

$$Q(\Theta \mid \Theta^{(t)}) = \mathbb{E}_{\Theta^{(t)}} \left[\log L_c(\Theta) \mid \mathbf{Y}_1, \dots, \mathbf{Y}_n \right],$$

which depends on conditional expectations of functions of U_i and V_{ij} given the observed data.

Under the Gamma–Gamma specification, these conditional moments admit explicit expressions but do not lead to closed-form updates for all parameters. In practice, they are evaluated numerically using stable fixed-point iterations, yielding a computationally reliable GEM procedure. In the M-step, parameters are updated by maximizing $Q(\Theta \mid \Theta^{(t)})$. In particular, the baseline intensity parameters λ_j admit the closed-form update

$$\lambda_j^{(t+1)} = \frac{\sum_{i=1}^{n} y_{ij}}{\sum_{i=1}^{n} \mathbb{E}_{\Theta^{(t)}} \left[U_i V_{ij} \mid \mathbf{Y}_i \right]}, \qquad j = 1, \dots, d. \tag{7}$$

The frailty parameters are updated through one-dimensional numerical optimization.

Each GEM iteration produces a non-decreasing observed-data log-likelihood, and under standard regularity conditions the algorithm converges to a stationary point of the likelihood (Wu, 1983). From a computational perspective, the cost of each iteration scales linearly with both the sample size n and the dimension d. The conditional expectations required in the E-step are independent across observations and components and can therefore be computed in parallel. In practice, the algorithm converges in a moderate number of iterations, and numerical experiments indicate approximately linear growth of computation time with dimension.

4. Simulation studies

This section investigates the finite-sample behavior of the proposed shared–idiosyncratic frailty model and compares it with commonly used alternatives for multivariate count data. The purpose of the simulation studies is threefold. First, we assess the accuracy and stability of likelihood-based estimation under correct model specification. Second, we examine the ability of competing models to recover dependence features beyond second-order moments, including rank-based and tail dependence measures. Third, we study robustness under misspecification of the latent dependence structure using likelihood-based risk criteria. Throughout, the goal is not to identify a universally superior model, but to clarify the regimes in which each formulation provides an adequate representation of dependence.

4.1. Simulation Design

As discussed in Sections 2.3 and 3, different latent-variable constructions may induce identical marginal means and covariances while exhibiting markedly different dependence structures. This motivates simulation criteria that go beyond moment recovery. Accordingly, model performance is evaluated using a combination of estimation accuracy, recovery of rank-based and extremal dependence features, and likelihood-based measures of global fit.

Unless otherwise stated, all simulation experiments consider a bivariate count vector (Y_{i1}, Y_{i2}) generated according to

$$Y_{ij} \mid (U_i, V_{ij}) \sim \text{Poisson}(\lambda_j U_i V_{ij}), \qquad j = 1, 2,$$

with $(\lambda_1, \lambda_2) = (2, 3)$. The idiosyncratic frailties follow $V_{ij} \sim \text{Gamma}(\beta_j, \beta_j)$ with $(\beta_1, \beta_2) = (3, 5)$, while the shared frailty is $U_i \sim \text{Gamma}(\alpha, \alpha)$. The dependence strength is governed by $\alpha \in \{0.5, 1, 5\}$, representing strong,

moderate, and weak dependence regimes, respectively. Under the Gamma(α, α) specification, the shared frailty satisfies $\mathrm{Var}(U) = 1/\alpha$, so that cross-sectional dependence decreases as α increases. Sample sizes are $n \in \{100, 200, 500\}$, and each configuration is replicated $R = 1000$ times.

Estimation accuracy is quantified using Monte Carlo bias and root mean squared error (RMSE). For a generic parameter θ, these are defined as

$$\mathrm{Bias}(\hat{\theta}) = \frac{1}{R}\sum_{r=1}^{R}(\hat{\theta}^{(r)} - \theta), \qquad \mathrm{RMSE}(\hat{\theta}) = \left\{\frac{1}{R}\sum_{r=1}^{R}(\hat{\theta}^{(r)} - \theta)^2\right\}^{1/2},$$

where $\hat{\theta}^{(r)}$ denotes the estimate obtained in replication r.

To assess recovery of dependence beyond second-order moments, we consider Kendall's tau. For each replication, the empirical Kendall's tau is computed as

$$\hat{\tau}^{\mathrm{emp}} = \mathbb{P}\left\{(Y_1 - Y_1')(Y_2 - Y_2') > 0\right\} - \mathbb{P}\left\{(Y_1 - Y_1')(Y_2 - Y_2') < 0\right\},$$

where (Y_1', Y_2') denotes an independent copy of (Y_1, Y_2). The model-implied Kendall's tau, denoted τ^{mod}, is approximated by Monte Carlo simulation from the fitted model. Performance is summarized through bias and RMSE of $\hat{\tau}^{\mathrm{emp}}$ relative to τ^{mod}.

Extremal dependence at finite levels is evaluated using the joint upper-tail co-occurrence probability

$$p_{0.95} = \mathbb{P}\left(Y_1 > q_{0.95,1},\, Y_2 > q_{0.95,2}\right),$$

where $q_{0.95,j}$ denotes the empirical 95th percentile of Y_j. Empirical estimates of $p_{0.95}$ are compared to their model-based counterparts obtained from simulated draws under the fitted models.

Finally, overall distributional adequacy is assessed via the average negative log-likelihood (ANLL),

$$\mathrm{ANLL} = -\frac{1}{n}\sum_{i=1}^{n}\log p_{\hat{\theta}}(Y_{i1}, Y_{i2}),$$

where $p_{\hat{\theta}}$ denotes the fitted joint probability mass function. Lower ANLL values indicate better global fit.

4.2. Simulation study 1: Finite-sample estimation under correct specification

We first assess the finite-sample behavior of the maximum likelihood estimator under correct specification of the shared–idiosyncratic frailty model. Estimation is carried out using the GEM algorithm described in Section 3.8. Convergence is declared when the relative change in successive parameter estimates falls below 10^{-6}.

Across all dependence regimes, estimation of the marginal intensities is stable, with RMSE decreasing as the sample size increases. Estimation of the dependence parameter α is reliable under strong and moderate dependence but deteriorates when dependence is weak, reflecting the limited information content of near-independence settings and the resulting flattening of the likelihood surface as $\mathrm{Var}(U) \to 0$. Recall that under the Gamma(α, α) specification, the shared frailty satisfies $\mathrm{Var}(U) = 1/\alpha$, so that smaller values of α correspond to stronger cross-sectional dependence. Table 1 and Figure 1 summarize the finite-sample performance of the maximum likelihood estimators under correct specification of the shared–idiosyncratic frailty model. Across all dependence regimes, the estimators of the marginal intensities λ_1 and λ_2 exhibit stable behavior, with root mean squared errors (RMSEs) decreasing as the sample size increases from $n = 100$ to $n = 500$. Biases are generally small across all settings, confirming the accuracy of the GEM-based estimation procedure.

The dependence parameter α is estimated reliably in scenarios with strong ($\alpha = 0.5$) or moderate ($\alpha = 1$) cross-sectional dependence. In these regimes, both the RMSE and bias decrease with increasing sample size, reflecting the increasing information content and curvature of the likelihood surface. In contrast, when dependence is weak ($\alpha = 5$), estimation of α becomes substantially less precise, as evidenced by the much larger RMSE values. This behavior is consistent with the near-flatness of the likelihood function when $\mathrm{Var}(U)$ is small, which limits information about the shared frailty. Nevertheless, even in this challenging scenario, RMSEs decline with larger sample sizes, indicating that the estimator is consistent.

Figure 1 provides a visual representation of these patterns. For the marginal parameters λ_1 and λ_2, RMSEs decrease monotonically with sample size, and estimation accuracy improves as dependence weakens (larger α), reflecting

Table 1
Bias, RMSE and convergence rate for different values of α and sample size n.

α	n	λ_1		λ_2		α		Conv. rate
		Bias	RMSE	Bias	RMSE	Bias	RMSE	
0.5	100	-0.0332	0.3434	-0.0479	0.4969	0.0248	0.1077	1
	200	0.0076	0.2642	0.0322	0.3885	0.0004	0.0608	1
	500	-0.0174	0.1424	-0.0169	0.2207	0.0029	0.0419	1
1.0	100	-0.0230	0.2975	-0.0156	0.4190	0.0558	0.2372	1
	200	0.0018	0.2009	0.0023	0.2821	0.0242	0.1486	1
	500	-0.0079	0.1385	0.0064	0.1760	0.0041	0.0838	1
5.0	100	0.0081	0.2058	0.0005	0.2516	0.8262	2.8278	1
	200	-0.0067	0.1528	0.0102	0.1949	0.4039	1.6190	1
	500	-0.0013	0.0910	0.0039	0.1211	0.0873	0.9123	1

Table 2
Bias in dependence recovery relative to empirical measures (model minus empirical).

		Proposed		Shared-Poisson		PLN	
α	n	$\Delta\tau$	$\Delta p_{0.95}$	$\Delta\tau$	$\Delta p_{0.95}$	$\Delta\tau$	$\Delta p_{0.95}$
0.5	100	-0.0061	-0.0004	0.0779	0.0078	-0.1362	0.0020
1.0	200	-0.0016	-0.0001	0.1022	0.0059	-0.0829	0.0018
5.0	500	-0.0009	0.0000	0.0778	0.0011	-0.0100	0.0004

reduced variability induced by the shared frailty. The pattern is reversed for the dependence parameter α, where weak dependence leads to substantially higher RMSEs, particularly for smaller samples, whereas strong dependence yields more accurate estimation as n grows. These patterns illustrate the relative stability of the estimators across dependence regimes.

Overall, these results demonstrate that the proposed GEM estimation procedure performs well in finite samples, particularly for the marginal intensity parameters, and provides reliable inference for the dependence parameter under moderate to strong cross-sectional association. The findings also underscore the intrinsic difficulty of estimating weak dependence in small samples, a limitation inherent to models with near-independent latent structure. Convergence to the MLE was achieved in all scenarios, as indicated by the convergence rates reported in Table 1.

4.3. Simulation study 2: Recovery of dependence beyond second-order moments

This simulation study evaluates the ability of competing models to recover dependence features that are not fully characterized by second-order moments. Although the proposed model, the shared-frailty Poisson model, and the Poisson–lognormal (PLN) model are calibrated to reproduce marginal means and empirical covariances, they impose distinct latent dependence structures, leading to potentially different behavior with respect to rank-based association and joint tail dependence.

Table 3 reports Monte Carlo averages of Kendall's τ and the upper tail co-occurrence probability $p_{0.95}$. Under strong and moderate dependence ($\alpha = 0.5$ and 1.0), the proposed model closely reproduces the empirical values of both measures across all sample sizes, with consistently small deviations. This agreement indicates that the model captures concordance and tail clustering beyond what is implied by covariance alone. In contrast, the PLN model systematically underestimates Kendall's τ in these regimes, reflecting attenuated rank dependence induced by the Gaussian latent structure despite matching second-order moments. The shared-frailty Poisson model exhibits

Table 3
Recovery of rank-based and tail dependence beyond second-order moments. Monte Carlo averages of Kendall's τ and tail co-occurrence probability $p_{0.95}$.

		Empirical		Proposed		Shared-Poisson		PLN	
α	n	τ	$p_{0.95}$	τ	$p_{0.95}$	τ	$p_{0.95}$	τ	$p_{0.95}$
0.5	100	0.5478	0.0187	0.5417	0.0183	0.6257	0.0265	0.4116	0.0207
	200	0.5486	0.0183	0.5408	0.0185	0.6313	0.0267	0.4119	0.0207
	500	0.5469	0.0182	0.5423	0.0182	0.6338	0.0269	0.4082	0.0202
1.0	100	0.4189	0.0130	0.4097	0.0129	0.5128	0.0188	0.3366	0.0153
	200	0.4165	0.0131	0.4149	0.0130	0.5187	0.0190	0.3336	0.0149
	500	0.4194	0.0133	0.4172	0.0131	0.5264	0.0194	0.3381	0.0152
5.0	100	0.1511	0.0051	0.1511	0.0047	0.2232	0.0058	0.1459	0.0053
	200	0.1555	0.0046	0.1487	0.0047	0.2232	0.0059	0.1445	0.0053
	500	0.1555	0.0049	0.1546	0.0049	0.2333	0.0060	0.1455	0.0053

the opposite tendency, substantially overestimating both τ and $p_{0.95}$, consistent with confounding between shared dependence and idiosyncratic heterogeneity.

To better visualize these systematic deviations, Figure 2 summarizes the corresponding biases. The figure displays the bias in Kendall's τ and tail co-occurrence probability across representative dependence regimes. The proposed model exhibits bias that is uniformly close to zero for both measures, with little variation across settings. By contrast, the shared-frailty Poisson model shows persistent positive bias, particularly for tail co-occurrence under stronger dependence, while the PLN model displays systematic negative bias in Kendall's τ, with biases shrinking toward zero as dependence weakens. Importantly, these patterns remain visible even when empirical covariances are well matched, underscoring that moment-based agreement does not ensure accurate recovery of nonlinear or extremal dependence.

Table 2 summarizes these findings numerically. The proposed model shows negligible bias in both Kendall's τ and $p_{0.95}$ across dependence regimes, with no systematic dependence on sample size. Under weak dependence ($\alpha = 5.0$), differences among models diminish, as expected; nevertheless, the shared-frailty Poisson model continues to overstate rank dependence, whereas the proposed model remains essentially unbiased.

Overall, the combined evidence from Tables 2–3 and Figure 2 highlights that models calibrated to identical first- and second-order moments may differ substantially in their representation of concordance and joint tail behavior. Accurately recovering such features requires an explicit separation of shared and idiosyncratic sources of variability, particularly in applications where extremal dependence is of substantive interest.

4.4. Simulation study 3: Likelihood-based model comparison

We assess the relative performance of the competing models using the average negative log-likelihood (ANLL), which serves as a likelihood-based measure of predictive risk. Table 4 reports Monte Carlo averages of ANLL, together with their corresponding standard deviations, under correct model specification. Figure 3 provides a graphical summary of these results across different sample sizes and dependence regimes.

When the data-generating mechanism follows the proposed Gamma–Gamma frailty structure, the proposed model consistently attains the smallest ANLL across all experimental settings. This finding indicates superior likelihood fit and confirms that the proposed formulation more accurately represents the underlying dependence pattern among repeated counts.

The magnitude of the performance differences varies systematically with the strength of dependence. Under strong dependence ($\alpha = 0.5$), the proposed method substantially outperforms both the shared-Poisson and Poisson–lognormal (PLN) models. In particular, the PLN model exhibits considerably larger ANLL values, suggesting heightened sensitivity to misspecification of the latent correlation structure. These results illustrate that likelihood-based criteria are especially informative in settings where dependence plays a dominant role, even when alternative models adequately recover marginal moments.

As dependence weakens ($\alpha = 1.0$ and 5.0), the discrepancies among competing methods gradually diminish. In these regimes, the joint distribution approaches near-independence, and multiple formulations yield comparable

Table 4
Likelihood-based model comparison under correct specification. Monte Carlo averages of the average negative log-likelihood (ANLL), with standard deviations in parentheses.

α	n	Proposed	Shared-Poisson	PLN
0.5	100	3.6425 (0.2359)	3.7383 (0.2673)	5.0676 (0.5125)
	200	3.6962 (0.1733)	3.8035 (0.2042)	5.1245 (0.3245)
	500	3.6896 (0.1063)	3.7984 (0.1218)	5.0932 (0.2586)
1.0	100	3.9559 (0.1995)	4.0432 (0.2283)	4.7197 (0.3008)
	200	3.9606 (0.1287)	4.0465 (0.1513)	4.7099 (0.2361)
	500	3.9660 (0.0873)	4.0596 (0.1000)	4.7074 (0.1437)
5.0	100	4.0469 (0.1386)	4.1183 (0.1619)	4.2507 (0.1748)
	200	4.0495 (0.0971)	4.1188 (0.1171)	4.2506 (0.1213)
	500	4.0432 (0.0598)	4.1173 (0.0718)	4.2417 (0.0784)

approximations. Nevertheless, the proposed model maintains a systematic advantage across all sample sizes, indicating greater robustness to variations in dependence strength.

Increasing sample size leads to reduced variability in ANLL, as reflected by the decreasing standard deviations reported in Table 4. However, the relative ranking of the models remains stable, suggesting that the observed differences are not attributable to sampling fluctuations. This stability is further illustrated in Figure 3, where separation between models is most evident under strong dependence and narrows as dependence weakens.

Overall, these results demonstrate that correct specification of the latent dependence structure yields substantial gains in likelihood-based performance, particularly in highly correlated settings. The proposed Gamma–Gamma frailty model achieves a favorable balance between flexibility and parsimony, resulting in improved predictive accuracy and more reliable inference under the assumed data-generating mechanism.

4.5. Simulation study 4: Robustness to misspecification of the latent structure

To assess robustness, we consider two forms of latent misspecification. Throughout this section, the notation "A → B" indicates that data are generated under model A and fitted using model B.

In the first scenario, data are generated from a Poisson–lognormal model and fitted using the proposed Gamma–Gamma frailty specification. In the second scenario, data are generated from a heavy-tailed shared frailty model and fitted using a Poisson–lognormal specification. More precisely, in the heavy-tailed scenario the shared frailty follows a Pareto distribution $U \sim \text{Pareto}(u_0, k), \quad P(U > u) = \left(\frac{u_0}{u}\right)^k, \quad u \geq u_0$, with parameters $u_0 = 1$ and $k = 1.5$. Conditional on U, the counts are generated according to $Y_j \mid U \sim \text{Poisson}(\lambda_j U), \quad j = 1, \ldots, d$, independently across components. This specification induces regularly varying tails in the latent multiplier and corresponds to the heavy-tailed frailty setting considered in Section 3 (Theorem 3), where tail dependence arises from a shared latent factor with power-law decay.

Table 5 and Figure 4 summarize the results in terms of average negative log-likelihood (ANLL), rank-based dependence distortion ($\Delta\tau$), and tail co-occurrence distortion ($\Delta p_{0.95}$).

In the first scenario (Poisson–lognormal → Proposed), the proposed Gamma–Gamma specification exhibits only a moderate increase in ANLL relative to correct specification, indicating limited loss of likelihood efficiency under Gaussian latent misspecification. As shown in Table 5, the distortion in Kendall's τ remains small across all sample sizes, with mean deviations close to zero and decreasing variability as n increases. Figure 4(a) further confirms that rank-based dependence is recovered with little systematic bias. Similarly, tail co-occurrence distortions remain negligible (Figure 4(b)), suggesting that the proposed model provides a stable approximation even when the true latent structure is lognormal rather than Gamma.

In contrast, the second scenario (Heavy-tailed frailty → Poisson–lognormal) reveals a markedly different behavior. The Poisson–lognormal model exhibits substantially larger ANLL values, reflecting a clear deterioration in likelihood fit when the latent dependence mechanism possesses regularly varying tails. Moreover, the consistently positive values of $\Delta\tau$ indicate systematic overestimation of rank dependence. This effect becomes more pronounced as the sample size increases, suggesting that the misspecification is structural rather than due to sampling variability. Intuitively,

Table 5
Robustness to latent misspecification. Mean (standard deviation) over $R = 1000$ Monte Carlo replications. $\Delta\tau$ and $\Delta p_{0.95}$ denote model minus empirical values.

Scenario	n	ANLL	$\Delta\tau$	$\Delta p_{0.95}$
Heavy-tail → Poisson–lognormal				
	100	8.948 (3.448)	0.216 (0.117)	-0.001 (0.009)
	200	8.844 (2.694)	0.250 (0.100)	-0.000 (0.008)
	500	9.127 (2.185)	0.269 (0.077)	0.002 (0.006)
Poisson–lognormal → Proposed				
	100	4.062 (0.148)	0.015 (0.063)	-0.002 (0.007)
	200	4.066 (0.095)	0.020 (0.045)	-0.001 (0.005)
	500	4.067 (0.060)	0.020 (0.029)	-0.001 (0.004)

the Gaussian latent structure attempts to compensate for the heavy-tailed shared variability by inflating correlation strength, thereby distorting dependence measures.

Despite this deterioration in likelihood and rank dependence recovery, distortions in tail co-occurrence probabilities remain comparatively small. This indicates that extreme co-occurrence frequencies are partly captured through increased variance in the fitted Gaussian latent factor, even though the underlying tail mechanism is misspecified.

Overall, these results highlight an asymmetry in robustness. The proposed Gamma–Gamma framework remains stable when the true latent distribution is light-tailed, whereas Gaussian latent models are considerably less robust when dependence arises from heavy-tailed shared heterogeneity. This finding is consistent with the theoretical analysis of Section 3, where heavy-tailed frailties generate stronger extremal dependence that cannot be reproduced by lognormal mixing. These findings suggest that robustness depends primarily on the tail behavior of the latent dependence mechanism rather than on marginal distributional assumptions alone.

4.6. Simulation study 5: Comparison with copula-based models

To further investigate the statistical properties of the proposed Gamma–Gamma shared–idiosyncratic frailty model, we compare its performance with several widely used copula-based constructions for multivariate discrete data. Copula models provide a flexible and widely adopted framework for constructing multivariate distributions with specified marginal distributions (Nelsen, 2006; Joe, 2014). In particular, they allow dependence structures to be modeled separately from the marginal distributions. For discrete responses, inference typically relies on distributional transforms or pseudo-likelihood techniques to address the non-uniqueness of copulas associated with discrete margins (Genest and Nešlehová, 2007; Panagiotelis, Czado and Joe, 2012a). Copula-based models with count margins, including negative binomial specifications, have been extensively studied and applied in the literature on multivariate discrete data (Joe et al., 2010; Nikoloulopoulos and Karlis, 2009; Nikoloulopoulos, 2013). While copula models provide a flexible representation of dependence, they do not explicitly encode a generative mechanism for the dependence structure. In contrast, the proposed frailty model introduces a hierarchical latent-variable representation in which dependence arises through multiplicative shared and idiosyncratic random effects. This formulation yields a natural interpretation of dependence in terms of latent heterogeneity and provides a structural probabilistic mechanism generating correlation among counts. The simulation study therefore examines whether this hierarchical representation can reproduce complex dependence patterns and how it compares with copula-based approaches.

We consider three representative copula constructions combined with negative binomial margins: a Gaussian copula capturing symmetric dependence, a Clayton copula emphasizing lower-tail association, and a Frank copula allowing moderate symmetric dependence without tail dominance. Let $F_j(y)$ denote the marginal cumulative distribution function of component j. The joint distribution under a copula model is defined as $P(Y_1 \le y_1, \dots, Y_d \le y_d) = C_\theta\big(F_1(y_1), \dots, F_d(y_d)\big)$, where C_θ denotes a parametric copula with dependence parameter θ. Model parameters are estimated using maximum pseudo-likelihood after applying the distributional transform to the discrete margins. The competing model is the proposed Gamma–Gamma shared–idiosyncratic frailty model introduced in Section 2. In this formulation the counts satisfy $Y_j \mid (U, V_j) \sim \text{Poisson}(\lambda_j U V_j)$, with latent variables $U \sim \text{Gamma}(\alpha, \alpha)$, $V_j \sim \text{Gamma}(\beta_j, \beta_j)$, assumed mutually independent. This hierarchical structure induces positive dependence among the counts through the shared frailty U, while allowing additional heterogeneity through the idiosyncratic frailties V_j.

Model parameters are estimated using the generalized EM algorithm described in Section 3. Two data-generating mechanisms are considered. In the first setting, data are generated from the proposed Gamma–Gamma shared–idiosyncratic frailty model in order to evaluate how well copula constructions recover dependence induced by multiplicative latent heterogeneity. In the second setting, data are generated from a Gaussian copula model with negative binomial margins to assess the robustness of the frailty model under misspecification of the latent dependence structure. We consider dimensions $d \in \{2, 5, 10\}$ and sample sizes $n \in \{200, 500\}$. For each simulated dataset, observations are randomly divided into a training set (80%) used for model estimation and a test set (20%) used for performance evaluation. Results are averaged over $R = 50$ Monte Carlo replications.

Model performance is evaluated using several complementary criteria. Dependence recovery is assessed by comparing Kendall's τ implied by simulated draws from the fitted model with the empirical dependence observed in the test data. Tail dependence is examined through the upper-tail co-occurrence probability $p_{0.95} = P(Y_i > q_{0.95,i}, Y_j > q_{0.95,j})$,

where $q_{0.95,j}$ denotes the empirical 95th percentile of component j. To further investigate extremal dependence behavior, we also evaluate tail calibration curves based on joint exceedance probabilities $P(Y_i > q_i, Y_j > q_j)$,

for quantile levels $q \in [0.90, 0.99]$.

Computational efficiency is assessed through the time required to fit each model. In addition, for datasets generated from the Gamma–Gamma shared–idiosyncratic frailty model, we examine recovery of the latent variance components by comparing the estimated variances of the shared and idiosyncratic frailties with their true values. Finally, we study how dependence recovery deteriorates with increasing dimension by analyzing the absolute bias in Kendall's τ as a function of d.

Table 6 summarizes the results of the comparison between the proposed frailty model and the copula-based approaches. When the data are generated from a copula model, the copula-based constructions naturally provide the most accurate recovery of Kendall's τ, reflecting correct specification of the dependence structure. Nevertheless, the proposed frailty model remains competitive and produces reasonable estimates of both dependence and tail co-occurrence probabilities, indicating that multiplicative latent heterogeneity can approximate a wide range of dependence patterns.

In contrast, when the data are generated from the Gamma–Gamma frailty model, the proposed approach clearly outperforms the copula-based models in recovering the dependence structure. In particular, the bias in Kendall's τ remains small across all dimensions, while the copula models exhibit increasingly large deviations as the dimension grows. Similar patterns are observed for the tail dependence measures, highlighting the ability of the hierarchical frailty representation to capture extremal dependence induced by latent multiplicative effects.

Table 7 further investigates the recovery of the latent variance components under the frailty data-generating mechanism. The results indicate that the variance of the shared frailty U is accurately estimated across all simulation settings. Estimation of the idiosyncratic frailty variance shows a mild upward bias for small dimensions, but improves substantially as the dimension increases. This behavior reflects improved identifiability of the hierarchical structure as the number of observed components grows.

Taken together, these results illustrate two important features of the proposed model. First, the hierarchical Gamma–Gamma frailty formulation provides a structurally interpretable mechanism for generating dependence in multivariate count data. Second, despite being motivated by latent heterogeneity rather than copula-based dependence, the model remains flexible enough to approximate alternative dependence structures while offering clear interpretability of the underlying sources of variability.

5. Application to Epileptic Seizure Counts

We illustrate the interpretability and practical utility of the proposed shared–idiosyncratic frailty model using a classical longitudinal dataset on epileptic seizure counts originally analyzed by Thall and Vail (1990). This dataset, available in the `MASS` package (Venables and Ripley, 2002), has become a benchmark for methodological developments in correlated and overdispersed count data. The study followed individuals with epilepsy and recorded the number of seizures experienced during four consecutive post-baseline observation periods of equal length.

Let $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3}, Y_{i4})^\top$ denote the vector of seizure counts for subject i, where Y_{ij} represents the number of seizures observed in period j, for $i = 1, \ldots, n$ and $j = 1, \ldots, 4$. The resulting data consist of repeated count measurements nested within subjects, exhibiting substantial heterogeneity and temporal dependence.

Table 6
Simulation results comparing copula-based models and the proposed frailty model. Entries report absolute bias in Kendall's τ and upper-tail probability ($|d\tau|/|dTail|$). The last column reports average fitting time in seconds.

Scenario	Model	$d = 2$		$d = 5$		$d = 10$		Time (s)
		$n = 200$	$n = 500$	$n = 200$	$n = 500$	$n = 200$	$n = 500$	
Copula data-generating mechanism								
	Clayton	0.0455 / 0.0063	0.0470 / 0.0037	0.1427 / 0.0061	0.1657 / 0.0060	0.1570 / 0.0049	0.1645 / 0.0057	0.10
	Frank	0.0342 / 0.0050	0.0267 / 0.0022	0.0436 / 0.0043	0.0652 / 0.0044	0.0572 / 0.0032	0.0637 / 0.0041	0.13
	Gaussian	0.0476 / 0.0019	0.0434 / 0.0007	0.0113 / 0.0006	0.0300 / 0.0007	0.0074 / 0.0009	0.0142 / 0.0000	0.16
	Frailty	0.1436 / 0.0029	0.1432 / 0.0006	0.0785 / 0.0005	0.0930 / 0.0007	0.0755 / 0.0009	0.0816 / 0.0002	81.67
Gamma–Gamma frailty data-generating mechanism								
	Clayton	0.0866 / 0.0070	0.0945 / 0.0096	0.2495 / 0.0104	0.2567 / 0.0112	0.3039 / 0.0106	0.2981 / 0.0122	0.20
	Frank	0.0324 / 0.0052	0.0352 / 0.0077	0.0974 / 0.0079	0.1094 / 0.0086	0.1442 / 0.0077	0.1341 / 0.0090	0.22
	Gaussian	0.0750 / 0.0019	0.0707 / 0.0043	0.0590 / 0.0024	0.0692 / 0.0032	0.0702 / 0.0002	0.0613 / 0.0016	0.19
	Frailty	0.0561 / 0.0002	0.0557 / 0.0025	0.0075 / 0.0002	0.0192 / 0.0006	0.0211 / 0.0023	0.0093 / 0.0009	82.56

Table 7
Recovery of latent variance components under the Gamma–Gamma frailty data-generating mechanism. Reported values are Monte Carlo averages of the true and estimated variances of the shared frailty U and the mean idiosyncratic frailty V_j.

		Shared frailty U		Idiosyncratic frailty V_j	
d	n	True	Estimated	True	Estimated
2	200	0.965	0.905	0.305	0.444
	500	0.985	0.889	0.310	0.457
5	200	1.033	1.003	0.257	0.305
	500	1.012	0.981	0.258	0.309
10	200	0.997	0.979	0.206	0.230
	500	1.013	0.997	0.207	0.229

Two salient empirical features motivate the use of latent-variable models. First, seizure counts display pronounced overdispersion, with empirical variances consistently exceeding corresponding means across periods. Such deviations from equidispersion are commonly attributed to unobserved heterogeneity and are typically accommodated through mixture or random-effects formulations (Cameron and Trivedi, 2013; Hilbe, 2011). Second, strong positive within-subject dependence is evident, reflecting persistent differences in underlying disease severity and treatment response.

To evaluate the contribution of the proposed model, we compare it with several widely used alternatives representing distinct strategies for handling latent heterogeneity. All competing models share a common marginal mean structure but differ in their stochastic representations of variability and dependence. As a baseline, we consider the negative binomial (NB) model, which arises from a Poisson–gamma mixture and captures overdispersion through a single dispersion parameter (Hilbe, 2011). Although this model accommodates marginal extra-Poisson variation, it does not explicitly account for serial dependence.

We further examine a shared-Poisson frailty model, in which a subject-specific latent effect acts multiplicatively on the Poisson intensity. Conditional on this frailty, observations are independent across periods, with marginal correlation induced by the shared component (Hougaard, 2000; Winkelmann, 2008). In addition, we consider the Poisson lognormal (PLN) model, which introduces Gaussian random effects on the log-intensity scale and allows for flexible dependence patterns and substantial deviation from equidispersion (Aitchison and Ho, 1989; Karlis and Xekalaki, 2005).

The proposed shared–idiosyncratic frailty model extends these formulations by decomposing latent variability into two distinct components: a shared frailty capturing persistent subject-level susceptibility and period-specific idiosyncratic effects accounting for transient fluctuations. Conditional on these latent variables, seizure counts follow independent Poisson distributions, while marginal variability separates naturally into common and period-dependent contributions.

Table 8 and Table 9 summarize parameter estimates and structural characteristics for all models. Estimated marginal intensities are broadly comparable across specifications, indicating that differences in performance are primarily driven by higher-order distributional features rather than mean structure. The NB model absorbs extra-Poisson variation

through a single dispersion parameter, whereas the shared frailty and PLN models introduce latent variance components of similar magnitude, reflecting substantial unobserved heterogeneity.

By contrast, the proposed model yields a more refined characterization of latent variability. The estimated shared frailty variance, $\mathrm{Var}(U) = 0.889$, indicates pronounced persistent heterogeneity across subjects, while the idiosyncratic variances $\mathrm{Var}(V_j)$ reveal heterogeneous period-specific contributions. Notably, several periods exhibit negligible additional latent variation, suggesting that most dependence is driven by stable subject-level factors. This decomposition facilitates direct interpretation of both marginal overdispersion and serial correlation.

The adequacy of the fitted marginal distributions is further illustrated in Figure 5, which compares observed and model-implied count distributions across periods. The proposed model provides a close approximation to the empirical distributions throughout the support, particularly in the tails, where competing models tend to under- or overestimate extreme counts. These improvements reflect the model's ability to accommodate both persistent and transient sources of variability.

The ability of competing models to reproduce the observed dependence structure is further examined in Figure 6, which displays pairwise Kendall's τ matrices and the corresponding approximation errors. The empirical dependence matrix exhibits substantial positive association across periods, reflecting persistent within-subject correlation in seizure activity. The shared-Poisson frailty model captures the overall direction of dependence but tends to oversmooth pairwise associations, leading to systematic underestimation for several period pairs. The PLN model provides greater flexibility but displays noticeable discrepancies in specific off-diagonal entries.

By contrast, the proposed shared–idiosyncratic frailty model closely reproduces the empirical dependence pattern across all period pairs. This improvement is reflected in the substantially smaller approximation errors shown in the lower panels of Figure 6. In particular, errors are uniformly attenuated relative to competing models, indicating that the proposed latent decomposition yields a more faithful representation of both global and pair-specific dependence. These findings corroborate the likelihood-based comparison results and highlight the advantages of explicitly separating shared and period-specific sources of variability.

Beyond improved goodness-of-fit, the proposed framework facilitates direct interpretation of latent heterogeneity, as illustrated in Figure 7. The left panel displays the empirical distribution of the estimated shared frailty components $\hat{U}_i$, normalized to satisfy $E(U) = 1$. The right-skewed shape indicates substantial heterogeneity in baseline seizure susceptibility across individuals, with a small proportion of subjects exhibiting markedly elevated risk. The middle panel of Figure 7 demonstrates a strong monotone relationship between the estimated shared frailty and the total seizure burden $S_i = \sum_j Y_{ij}$. This association confirms that the shared component effectively captures persistent individual-level intensity, thereby providing a parsimonious summary of overall disease severity. The right panel summarizes idiosyncratic heterogeneity across observation periods after accounting for the shared frailty effect. While most periods exhibit modest residual variation, several display systematic departures from unity, indicating period-specific deviations not explained by global susceptibility alone. These patterns suggest the presence of transient influences, such as treatment effects or temporal fluctuations in disease activity, which are accommodated through the idiosyncratic latent components.

Taken together, Figures 5, 6, and 7 demonstrate that the proposed model achieves three complementary objectives: accurate reproduction of marginal distributions, faithful recovery of dependence structure, and transparent interpretation of latent heterogeneity. This combination distinguishes the proposed framework from existing alternatives, which typically prioritize flexibility or parsimony at the expense of interpretability or identifiability.

Model comparison results reported in Table 10 further support these findings. The proposed specification attains the lowest average negative log-likelihood, indicating superior overall fit relative to the shared-Poisson and PLN alternatives. While competing models perform adequately under specific criteria, the proposed framework achieves a balanced improvement across likelihood-based and dependence-sensitive measures. These results highlight the inferential benefits of separating persistent and transient sources of heterogeneity in longitudinal count data.

Table 8
Estimated marginal intensities and dispersion parameters for competing models.

Model	λ_{Y_1}	λ_{Y_2}	λ_{Y_3}	λ_{Y_4}	Dispersion / model parameters
NB	8.95	8.36	8.44	7.31	$\theta = 0.903$
Shared-Poisson	8.95	8.36	8.44	7.30	$\alpha = 1.109$
PLN	7.88	7.36	7.44	6.43	$\sigma = 0.833$
Proposed	8.83	8.79	7.95	7.51	$\alpha = 1.125$
					$\boldsymbol{\beta} = (5.135,\ 20.085,\ 3.677,\ 5524.597)$

Table 9
Latent variance structure and dependence mechanisms for competing models.

Model	Latent variance	Dependence mechanism	Interpretation
NB	–	–	Marginal overdispersion
Shared-Poisson	$\mathrm{Var}(U) = 0.902$	Shared frailty	Patient-level heterogeneity
PLN	$\mathrm{Var}(U) = 1.001$	Lognormal latent factor	Flexible dependence structure
Proposed	$\mathrm{Var}(U) = 0.889,\ \mathrm{Var}(V_j) = (0.195,\ 0.05,\ 0.272,\ 0)$	Shared + idiosyncratic	Variance decomposition

Table 10
Model comparison and goodness-of-fit diagnostics.

	Likelihood	Dependence	Tail behavior
Model	ANLL (↓)	RMSE_τ (↓)	$\mathrm{RMSE}_{\mathrm{tail}}$ (↓)
Shared-Poisson	11.895	0.173	**0.012**
PLN	11.790	**0.111**	0.021
Proposed	**11.017**	0.182	0.014

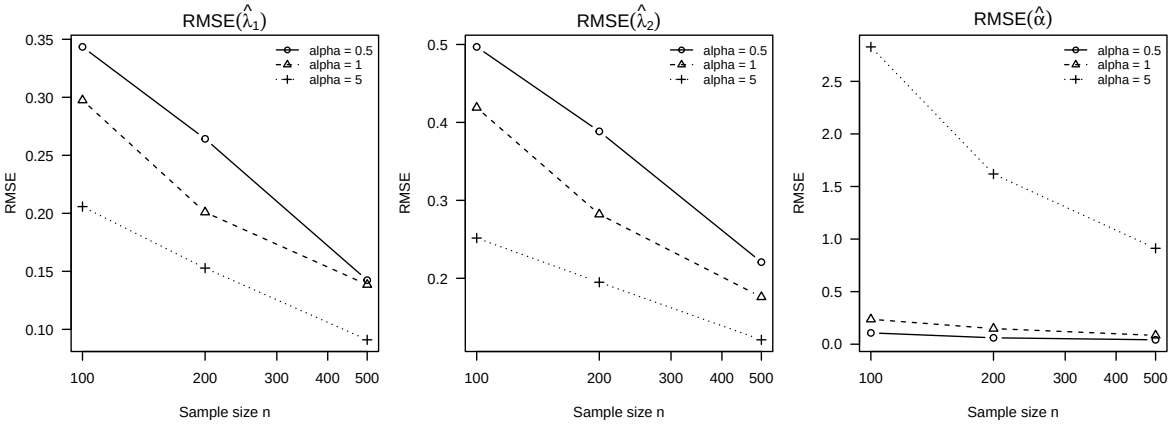

Figure 1: Monte Carlo RMSE of the MLEs under correct model specification.

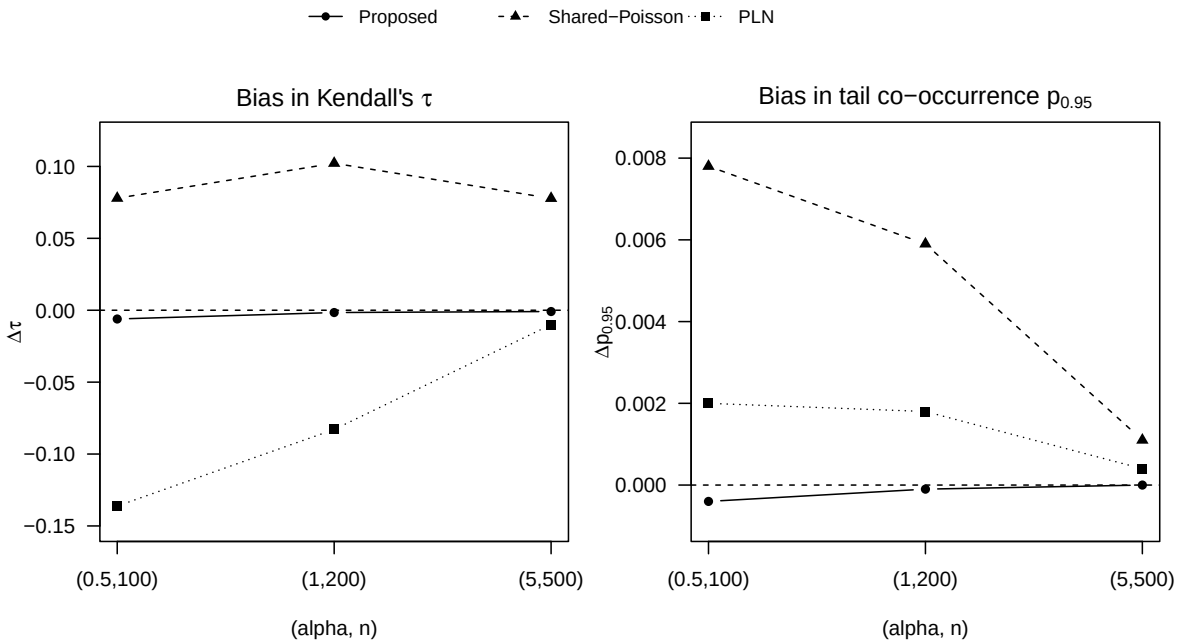

Figure 2: Bias in recovery of rank-based and tail dependence

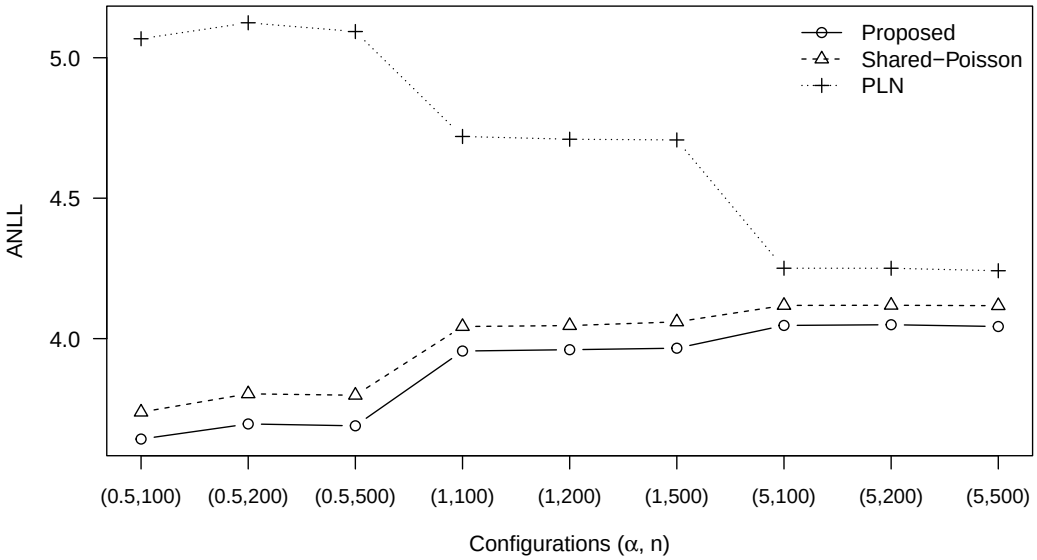

Figure 3: Likelihood-based model comparison (ANLL)

Figure 4: Robustness under latent misspecification. (a) Model minus empirical Kendall's τ. (b) Model minus empirical tail co-occurrence probability $p_{0.95}$. Heavy-tailed data are fitted with the Poisson–lognormal model, whereas Poisson–lognormal data are fitted with the proposed Gamma–Gamma specification. Results are shown across sample sizes.

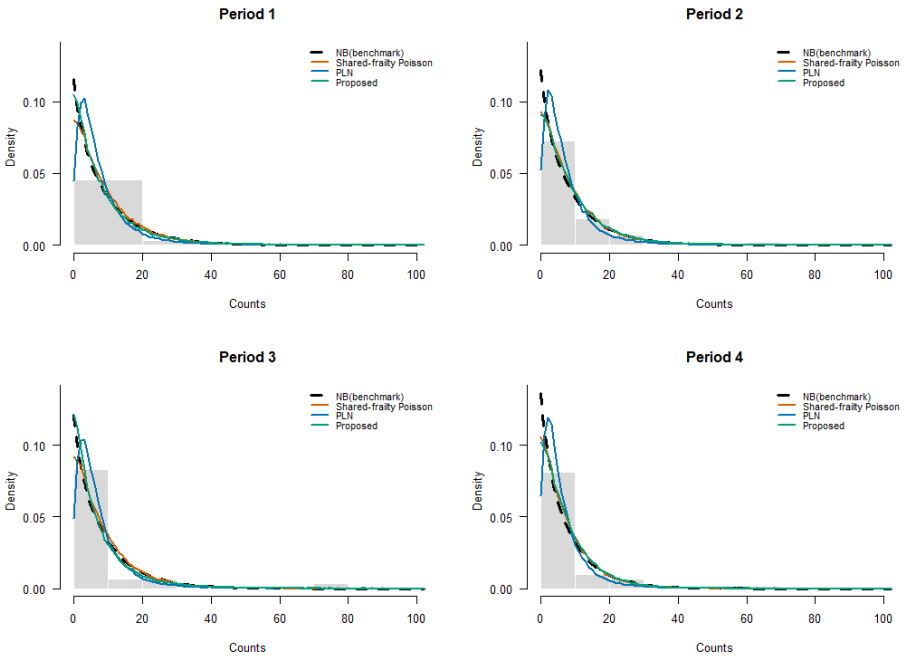

Figure 5: Marginal distribution fits across periods for the epilepsy data.

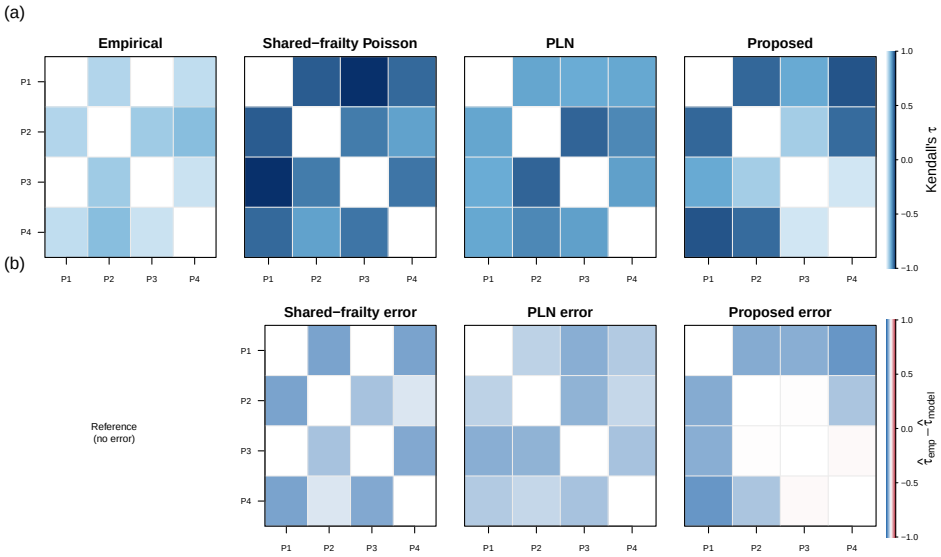

Figure 6: Estimated dependence structure and associated approximation errors. (a) Pairwise dependence matrices based on Kendall's τ, showing the empirical estimate and the corresponding model-based estimates for the competing models. (b) Approximation errors defined as the difference between empirical and model-based dependence (empirical − model). The empirical dependence serves as the reference and therefore has no associated error panel.

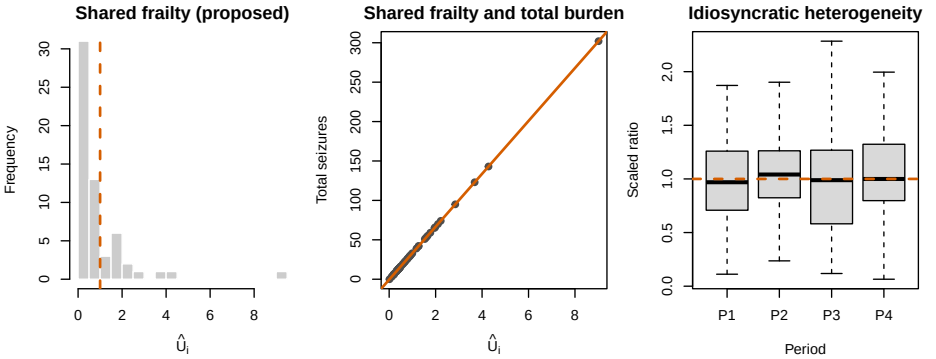

Figure 7: Latent interpretation under the proposed shared–idiosyncratic frailty model. Left: distribution of the estimated shared frailty $\hat{U}_i$, representing subject-specific global susceptibility, with the dashed line indicating the unit-mean normalization $E(U) = 1$. Middle: relationship between the estimated shared frailty and the total seizure burden $S_i = \sum_j Y_{ij}$, illustrating that the shared component captures overall individual intensity. Right: idiosyncratic heterogeneity across periods, summarized by scaled ratios approximating period-specific deviations after accounting for the shared frailty effect; the horizontal dashed line at 1 corresponds to perfect agreement with the shared component alone.

6. Conclusion

This paper develops a shared–idiosyncratic frailty framework for multivariate count data that separates cross-sectional dependence from outcome-specific heterogeneity through an explicit latent decomposition. The proposed construction retains mixed-Poisson marginal behavior while inducing dependence solely through a shared latent multiplier, yielding parameters with direct probabilistic interpretation. Tractable characterization of the joint distribution follows from the probability generating function, leading to closed-form expressions for moments and cross-covariances and to identifiability under mild normalization constraints.

We also propose likelihood-based inference through an EM algorithm that leverages the conditional independence implied by the latent representation. A suite of simulation studies clarifies the finite-sample and modeling implications of the decomposition. Under correct specification, the GEM-based estimators exhibit stable behavior for marginal intensity parameters across sample sizes, with the dependence parameter becoming more difficult to estimate as the dependence weakens, in line with the reduced curvature of the likelihood surface near independence. Importantly, simulations further show that models calibrated to match first- and second-order moments may differ substantially in their representation of nonlinear dependence. In particular, the proposed framework more accurately recovers rank-based and finite-level extremal dependence summaries, as assessed by Kendall's τ and the joint tail co-occurrence probability $p_{0.95}$, highlighting that covariance agreement alone does not guarantee adequate recovery of concordance or tail clustering. Likelihood risk comparisons based on ANLL reinforce these findings: when the latent structure is correctly specified, the proposed model achieves systematic improvements, with gains most pronounced under strong dependence. Robustness experiments under latent misspecification indicate that performance is asymmetric across modeling families, with the proposed Gamma–Gamma specification exhibiting relatively stable likelihood and rank-based dependence recovery when the true latent structure is lognormal, whereas Gaussian latent specifications can deteriorate under heavy-tailed shared heterogeneity.

The empirical analysis of the epileptic seizure data illustrates the practical relevance of these properties in a longitudinal setting with both marginal overdispersion and within-subject association. The proposed model reproduces marginal distributional features across periods, closely matches the observed dependence matrix, and yields interpretable latent summaries: the estimated shared frailty captures persistent subject-level susceptibility and aligns with overall seizure burden, while the idiosyncratic components reveal period-specific deviations beyond the shared severity effect. Together, these results demonstrate how separating shared and idiosyncratic sources of variability can improve both fit and interpretability in correlated count data.

Several extensions are worth pursuing. First, richer intensity models incorporating covariates and offsets could allow shared and outcome-specific regression effects, including interactions. Second, frailty distributions may be generalized beyond parametric Gamma families to semiparametric or nonparametric mixing distributions, potentially improving robustness while preserving identifiability via Laplace transform arguments. Third, the framework may be adapted to accommodate missing outcomes, informative observation windows, and longitudinal follow-up with time-varying intensities. Finally, dependence-focused diagnostics and model assessment tools particularly those targeting joint tail behavior would further strengthen applied deployment.

Overall, the shared–idiosyncratic frailty decomposition provides a principled and interpretable route to modeling multivariate counts, bridging identifiable latent structure with tractable likelihood-based inference and empirical recovery of dependence features beyond second-order moments.

Author contributions

Conceptualization: E. A., A. F. L., A. S. and S. M. M.; Methodology: E. A., A. F. L., A. S. and S. M. M.; Software: E. A., A. F. L., A. S. and S. M. M.; Formal Analysis: E. A., A. F. L., A. S. and S. M. M.; Resources: E. A., A. F. L., A. S. and S. M. M.; Writing—Original Draft Preparation: E. A., A. F. L., A. S. and S. M. M.; Review and Editing: E. A., A. F. L., A. S. and S. M. M.; Visualization: E. A., A. F. L., A. S. and S. M. M.

Financial disclosure

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Conflict of interest

The authors declare no potential conflict of interests.

Appendix A. Proofs of distributional and dependence results

Proof of Proposition 1. Fix $j \in \{1, \dots, d\}$. By the model construction, conditional on $(U, V_j) = (u, v)$ we have

$$Y_j \mid (U = u, V_j = v) \sim \text{Poisson}(\lambda_j uv),$$

so that, for any $y \in \mathbb{N}_0$,

$$\mathbb{P}(Y_j = y \mid U = u, V_j = v) = \frac{e^{-\lambda_j uv}(\lambda_j uv)^y}{y!}. \tag{8}$$

Since U and V_j are independent with respective laws G_0 and G_j on $(0, \infty)$, the joint law of (U, V_j) is the product measure $G_0 \otimes G_j$. Therefore, by the law of total probability,

$$\begin{aligned}\mathbb{P}(Y_j = y) &= \int_0^\infty \int_0^\infty \mathbb{P}(Y_j = y \mid U = u, V_j = v)\, dG_j(v)\, dG_0(u) \\ &= \int_0^\infty \int_0^\infty \frac{e^{-\lambda_j uv}(\lambda_j uv)^y}{y!}\, dG_j(v)\, dG_0(u).\end{aligned} \tag{9}$$

This integral representation is precisely a Poisson mixture: if we define the nonnegative random variable

$$\Lambda_j := \lambda_j U V_j,$$

then (9) can be rewritten as

$$\mathbb{P}(Y_j = y) = \mathbb{E}\left[\frac{e^{-\Lambda_j}\Lambda_j^y}{y!}\right], \qquad y \in \mathbb{N}_0,$$

which is the defining form of a mixed Poisson distribution with mixing variable Λ_j. This proves the proposition.

For completeness, we also record the marginal probability generating function (pgf). For any $s \in [0, 1]$, using (8) and Tonelli's theorem (the integrand is nonnegative), we obtain

$$\begin{aligned}\mathbb{E}\left[s^{Y_j}\right] &= \int_0^\infty \int_0^\infty \sum_{y=0}^\infty s^y \frac{e^{-\lambda_j uv}(\lambda_j uv)^y}{y!}\, dG_j(v)\, dG_0(u) \\ &= \int_0^\infty \int_0^\infty e^{-\lambda_j uv} \exp\left(\lambda_j uv\, s\right) dG_j(v)\, dG_0(u) \\ &= \int_0^\infty \int_0^\infty \exp\left(\lambda_j uv(s-1)\right) dG_j(v)\, dG_0(u) = \mathbb{E}\left[\exp\left(\Lambda_j(s-1)\right)\right].\end{aligned}$$

This is the pgf of a mixed Poisson variable, and it further confirms that Y_j is mixed Poisson with mixing variable $\Lambda_j = \lambda_j U V_j$. □

Proof of Proposition 2. Fix $j \neq k$. Under the shared–idiosyncratic frailty model,

$$Y_j \mid (U, V_j) \sim \text{Poisson}(\lambda_j U V_j), \qquad Y_k \mid (U, V_k) \sim \text{Poisson}(\lambda_k U V_k),$$

and the components of $\mathbf{Y}$ are conditionally independent given $(U, \mathbf{V}) = (U, V_1, \dots, V_d)$.

We first compute the covariance between Y_j and Y_k. By the law of total covariance,

$$\text{Cov}(Y_j, Y_k) = \mathbb{E}\left[\text{Cov}(Y_j, Y_k \mid U, \mathbf{V})\right] + \text{Cov}\left(\mathbb{E}[Y_j \mid U, \mathbf{V}], \mathbb{E}[Y_k \mid U, \mathbf{V}]\right).$$

Because Y_j and Y_k are conditionally independent given $(U, \mathbf{V})$, the first term is zero:

$$\mathrm{Cov}(Y_j, Y_k \mid U, \mathbf{V}) = 0 \qquad \text{a.s.}$$

Hence,

$$\mathrm{Cov}(Y_j, Y_k) = \mathrm{Cov}\big(\mathbb{E}[Y_j \mid U, \mathbf{V}], \mathbb{E}[Y_k \mid U, \mathbf{V}]\big).$$

Now, using the conditional Poisson specification,

$$\mathbb{E}[Y_j \mid U, \mathbf{V}] = \lambda_j U V_j, \qquad \mathbb{E}[Y_k \mid U, \mathbf{V}] = \lambda_k U V_k.$$

Therefore,

$$\mathrm{Cov}(Y_j, Y_k) = \lambda_j \lambda_k \,\mathrm{Cov}(UV_j, UV_k).$$

Since U, V_j, V_k are mutually independent, we obtain

$$\mathbb{E}[UV_j] = \mathbb{E}[U]\mathbb{E}[V_j], \qquad \mathbb{E}[UV_k] = \mathbb{E}[U]\mathbb{E}[V_k],$$

and

$$\mathbb{E}[U^2 V_j V_k] = \mathbb{E}[U^2]\mathbb{E}[V_j]\mathbb{E}[V_k].$$

Thus,

$$\begin{aligned}
\mathrm{Cov}(UV_j, UV_k) &= \mathbb{E}[U^2 V_j V_k] - \mathbb{E}[UV_j]\mathbb{E}[UV_k] \\
&= \mathbb{E}[U^2]\mathbb{E}[V_j]\mathbb{E}[V_k] - \mathbb{E}[U]^2\mathbb{E}[V_j]\mathbb{E}[V_k] \\
&= \left(\mathbb{E}[U^2] - \mathbb{E}[U]^2\right)\mathbb{E}[V_j]\mathbb{E}[V_k] \\
&= \mathrm{Var}(U)\,\mathbb{E}[V_j]\mathbb{E}[V_k].
\end{aligned}$$

It follows that

$$\mathrm{Cov}(Y_j, Y_k) = \lambda_j \lambda_k \,\mathrm{Var}(U)\,\mathbb{E}[V_j]\mathbb{E}[V_k].$$

Under the normalization $\mathbb{E}(U) = 1$ and $\mathbb{E}(V_j) = 1$ for all j, this simplifies to

$$\mathrm{Cov}(Y_j, Y_k) = \lambda_j \lambda_k \,\mathrm{Var}(U).$$

We next derive the correlation formula. By definition,

$$\mathrm{Corr}(Y_j, Y_k) = \frac{\mathrm{Cov}(Y_j, Y_k)}{\sqrt{\mathrm{Var}(Y_j)\mathrm{Var}(Y_k)}}.$$

Substituting the covariance expression just obtained gives

$$\mathrm{Corr}(Y_j, Y_k) = \frac{\lambda_j \lambda_k \,\mathrm{Var}(U)}{\sqrt{\mathrm{Var}(Y_j)\mathrm{Var}(Y_k)}}.$$

Finally, since $\lambda_j, \lambda_k > 0$ and $\mathrm{Var}(Y_j), \mathrm{Var}(Y_k) > 0$, we have

$$\mathrm{Corr}(Y_j, Y_k) = 0 \quad \Longleftrightarrow \quad \mathrm{Cov}(Y_j, Y_k) = 0 \quad \Longleftrightarrow \quad \mathrm{Var}(U) = 0.$$

Equivalently, cross-sectional dependence vanishes if and only if the shared frailty is degenerate almost surely at its mean. This completes the proof. □

Proof of Theorem 1. We apply the law of total covariance:

$$\mathrm{Cov}(Y_j, Y_k) = \mathbb{E}\big[\mathrm{Cov}(Y_j, Y_k \mid U, \mathbf{V})\big] + \mathrm{Cov}\big(\mathbb{E}[Y_j \mid U, V_j], \mathbb{E}[Y_k \mid U, V_k]\big).$$

Conditional on $(U, \mathbf{V})$, the components of $\mathbf{Y}$ are independent, so that

$$\mathrm{Cov}(Y_j, Y_k \mid U, \mathbf{V}) = 0.$$

Moreover,

$$\mathbb{E}[Y_j \mid U, V_j] = \lambda_j U V_j, \qquad \mathbb{E}[Y_k \mid U, V_k] = \lambda_k U V_k.$$

Since V_j and V_k are independent and independent of U, it follows that

$$\mathrm{Cov}\big(\mathbb{E}[Y_j \mid U, V_j], \mathbb{E}[Y_k \mid U, V_k]\big) = \lambda_j \lambda_k \mathrm{Var}(U)\mathbb{E}[V_j]\mathbb{E}[V_k].$$

Combining these terms yields the stated covariance expression. □

Proof of Proposition 3. Recall that the multivariate probability generating function (pgf) of $\mathbf{Y} = (Y_1, \dots, Y_d)^\top$ is defined, for $\mathbf{s} = (s_1, \dots, s_d) \in [0,1]^d$, by

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\prod_{j=1}^{d} s_j^{Y_j}\right].$$

We compute this quantity by conditioning on the latent frailties $(U, \mathbf{V}) = (U, V_1, \dots, V_d)$. By the tower property,

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\mathbb{E}\left(\prod_{j=1}^{d} s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right)\right]. \tag{10}$$

Under the proposed model, conditional on $(U, \mathbf{V})$, the components $Y_1, \dots, Y_d$ are independent and

$$Y_j \mid (U, \mathbf{V}) \sim \mathrm{Poisson}(\lambda_j U V_j), \qquad j = 1, \dots, d.$$

Hence the conditional expectation in (10) factorizes:

$$\mathbb{E}\left(\prod_{j=1}^{d} s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right) = \prod_{j=1}^{d} \mathbb{E}\left(s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right) = \prod_{j=1}^{d} \mathbb{E}\left(s_j^{Y_j} \,\middle|\, U, V_j\right). \tag{11}$$

It remains to compute the pgf of a Poisson random variable. If $Z \sim \mathrm{Poisson}(\mu)$, then for $s \in [0,1]$,

$$\mathbb{E}[s^Z] = \sum_{y=0}^{\infty} s^y \frac{e^{-\mu}\mu^y}{y!} = e^{-\mu} \sum_{y=0}^{\infty} \frac{(\mu s)^y}{y!} = e^{-\mu} e^{\mu s} = \exp\{\mu(s-1)\}. \tag{12}$$

Applying (12) with $\mu = \lambda_j U V_j$ yields

$$\mathbb{E}\left(s_j^{Y_j} \,\middle|\, U, V_j\right) = \exp\left\{\lambda_j U V_j (s_j - 1)\right\}.$$

Substituting into (11) gives

$$\mathbb{E}\left(\prod_{j=1}^{d} s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right) = \prod_{j=1}^{d} \exp\left\{\lambda_j U V_j (s_j - 1)\right\} = \exp\left\{U \sum_{j=1}^{d} \lambda_j V_j (s_j - 1)\right\}.$$

Finally, taking expectation with respect to $(U, \mathbf{V})$ in (10) yields

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\exp\left\{U \sum_{j=1}^{d} \lambda_j V_j (s_j - 1)\right\}\right], \qquad \mathbf{s} \in [0,1]^d,$$

which is the desired expression. □

Proof of Proposition 4. Recall that the joint probability generating function of $\mathbf{Y}$ is defined by

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\prod_{j=1}^{d} s_j^{Y_j}\right], \qquad \mathbf{s} = (s_1, \dots, s_d) \in [0,1]^d.$$

We derive the stated representation by repeated conditioning on the latent frailties.

Let $\mathbf{V} = (V_1, \dots, V_d)$. By the tower property,

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\mathbb{E}\left(\prod_{j=1}^{d} s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right)\right].$$

Conditional on $(U, \mathbf{V})$, the random variables $Y_1, \dots, Y_d$ are independent, and

$$Y_j \mid (U, \mathbf{V}) \sim \text{Poisson}(\lambda_j U V_j), \qquad j = 1, \dots, d.$$

Therefore,

$$\mathbb{E}\left(\prod_{j=1}^{d} s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right) = \prod_{j=1}^{d} \mathbb{E}\left(s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right).$$

Now, if $Y \sim \text{Poisson}(\mu)$, its pgf is

$$\mathbb{E}(s^Y) = \exp\{\mu(s-1)\}, \qquad s \in [0,1].$$

Applying this with $\mu = \lambda_j U V_j$ yields

$$\mathbb{E}\left(s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right) = \exp\left\{\lambda_j U V_j (s_j - 1)\right\} = \exp\left\{-\lambda_j U V_j (1 - s_j)\right\}.$$

Hence,

$$\mathbb{E}\left(\prod_{j=1}^{d} s_j^{Y_j} \,\middle|\, U, \mathbf{V}\right) = \prod_{j=1}^{d} \exp\left\{-\lambda_j U V_j (1 - s_j)\right\}.$$

Substituting this into the tower property gives

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\prod_{j=1}^{d} \exp\left\{-\lambda_j U V_j (1 - s_j)\right\}\right].$$

We next integrate out the idiosyncratic frailties conditionally on U. Using the tower property again,

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\mathbb{E}\left(\prod_{j=1}^{d} \exp\left\{-\lambda_j U V_j (1 - s_j)\right\} \,\middle|\, U\right)\right].$$

Conditional on U, the variables $V_1, \dots, V_d$ remain independent, and thus

$$\mathbb{E}\left(\prod_{j=1}^{d} \exp\left\{-\lambda_j U V_j (1 - s_j)\right\} \,\middle|\, U\right) = \prod_{j=1}^{d} \mathbb{E}\left(\exp\left\{-\lambda_j U V_j (1 - s_j)\right\} \,\middle|\, U\right).$$

Since U is fixed inside the conditional expectation and V_j is independent of U, we obtain

$$\mathbb{E}\left(\exp\left\{-\lambda_j U V_j (1 - s_j)\right\} \,\middle|\, U\right) = \mathbb{E}\left[\exp\left\{-\lambda_j U (1 - s_j) V_j\right\}\right] = L_j\left(\lambda_j U (1 - s_j)\right).$$

Therefore,

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\prod_{j=1}^{d} L_j\big(\lambda_j U(1-s_j)\big)\right],$$

which proves the first representation.

To derive the integral form, note that the outer expectation above is taken with respect to the distribution of the shared frailty U, say G_0. Hence,

$$P_{\mathbf{Y}}(\mathbf{s}) = \int_0^\infty \prod_{j=1}^{d} L_j\big(\lambda_j u(1-s_j)\big)\, dG_0(u),$$

which is precisely the second representation. □

Proof of Theorem 2. We prove that $Y = (Y_1, \dots, Y_d)^\top$ is associated, i.e., for any coordinatewise non-decreasing measurable functions f and g such that the covariance exists, $\mathrm{Cov}(f(Y), g(Y)) \geq 0$.

First, conditional on $(U, \mathbf{V}) = (u, v_1, \dots, v_d)$, the components $Y_1, \dots, Y_d$ are independent and satisfy

$$Y_j \mid (U = u, V_j = v_j) \sim \mathrm{Poisson}(\lambda_j u v_j), \qquad j = 1, \dots, d.$$

Independence implies association (see Esary et al., 1967). Hence, for each fixed $(u, \mathbf{v})$ and any coordinatewise non-decreasing functions f and g,

$$\mathrm{Cov}(f(Y), g(Y) \mid U = u, \mathbf{V} = \mathbf{v}) \geq 0. \tag{13}$$

Next, fix $\mathbf{v} = (v_1, \dots, v_d)$. For $u > 0$, let $\mathbb{P}_u^{\mathbf{v}}$ denote the conditional law of Y given $(U = u, \mathbf{V} = \mathbf{v})$, that is, the product of independent Poisson distributions with means $\lambda_j u v_j$. For each j, the family $\{\mathrm{Poisson}(\lambda_j u v_j) : u > 0\}$ is stochastically increasing in u. Consequently, for any bounded coordinatewise non-decreasing function φ, the map

$$u \longmapsto \mathbb{E}_u^{\mathbf{v}}[\varphi(Y)]$$

is non-decreasing. Equivalently, the kernel $u \mapsto \mathbb{P}_u^{\mathbf{v}}$ is stochastically increasing with respect to the usual product order on $\mathbb{N}_0^d$ (see, e.g., Shaked and Shanthikumar, 2007, Section 1.7).

Finally, the unconditional law of Y can be written as the mixture

$$\mathbb{P}(Y \in \cdot) = \int \mathbb{P}_u^{\mathbf{V}}(\cdot)\, dG_0(u),$$

where $\mathbf{V}$ is independent of U and G_0 is the law of U. By (13), each conditional distribution $\mathbb{P}_u^{\mathbf{v}}$ is associated, and the family $\{\mathbb{P}_u^{\mathbf{v}} : u > 0\}$ is stochastically increasing in u. A closure property of association (Theorem 2.1 in Esary et al., 1967) then implies that the mixture distribution is also associated.

This proves that Y is associated. □

Appendix B. Proof of identifiability

Proof of Theorem 4.

Let $\Theta = (\lambda_1, \dots, \lambda_d, G_0, G_1, \dots, G_d)$ and $\widetilde{\Theta} = (\tilde{\lambda}_1, \dots, \tilde{\lambda}_d, \tilde{G}_0, \tilde{G}_1, \dots, \tilde{G}_d)$ be two parameter sets that induce the same law for $\mathbf{Y} = (Y_1, \dots, Y_d)^\top$. We show that $\Theta = \widetilde{\Theta}$ under the normalization $\mathbb{E}[U] = 1$ and $\mathbb{E}[V_j] = 1$ for all j.

First, equality of the joint probability generating functions implies equality of the associated multivariate Laplace transforms. By Proposition 3, for all $\mathbf{s} \in [0,1]^d$,

$$P_{\mathbf{Y}}(\mathbf{s}) = \mathbb{E}\left[\exp\left\{U \sum_{j=1}^{d} \lambda_j V_j (s_j - 1)\right\}\right] = \mathbb{E}\left[\exp\left\{\tilde{U} \sum_{j=1}^{d} \tilde{\lambda}_j \tilde{V}_j (s_j - 1)\right\}\right].$$

Setting $t_j = 1 - s_j$ and writing $\mathbf{t} = (t_1, \dots, t_d)$ yields, for all $\mathbf{t} \in [0,1]^d$,

$$\mathbb{E}\left[\exp\left(-\sum_{j=1}^{d} t_j\, \lambda_j U V_j\right)\right] = \mathbb{E}\left[\exp\left(-\sum_{j=1}^{d} t_j\, \tilde{\lambda}_j \tilde{U} \tilde{V}_j\right)\right]. \tag{14}$$

Hence the joint Laplace transforms of the nonnegative random vectors

$$X := (X_1, \dots, X_d)^\top, \qquad X_j := \lambda_j U V_j,$$

and

$$\tilde{X} := (\tilde{X}_1, \dots, \tilde{X}_d)^\top, \qquad \tilde{X}_j := \tilde{\lambda}_j \tilde{U} \tilde{V}_j,$$

coincide on $[0,1]^d$. By analyticity of Laplace transforms for nonnegative random vectors, this equality extends to a neighborhood of the origin in $\mathbb{R}_+^d$. By uniqueness of the multivariate Laplace transform on $(0,\infty)^d$, it follows that

$$X \overset{d}{=} \tilde{X}. \tag{15}$$

Taking expectations in (15) allows us to identify the marginal intensity parameters. Since $X_j = \lambda_j U V_j$ and U is independent of V_j,

$$\mathbb{E}[X_j] = \lambda_j \mathbb{E}[U]\mathbb{E}[V_j] = \lambda_j \quad \text{under } \mathbb{E}[U] = \mathbb{E}[V_j] = 1,$$

and similarly $\mathbb{E}[\tilde{X}_j] = \tilde{\lambda}_j$. Hence

$$\lambda_j = \tilde{\lambda}_j, \qquad j = 1, \dots, d.$$

In the remainder, we therefore assume $\lambda_j = \tilde{\lambda}_j$ and focus on the identification of the mixing distributions.

With $\lambda_j = \tilde{\lambda}_j$, identity (14) reduces to

$$\mathbb{E}\left[\exp\left(-U \sum_{j=1}^{d} t_j\, \lambda_j V_j\right)\right] = \mathbb{E}\left[\exp\left(-\tilde{U} \sum_{j=1}^{d} t_j\, \lambda_j \tilde{V}_j\right)\right], \qquad \mathbf{t} \approx \mathbf{0}. \tag{16}$$

Let $L_U(a) = \mathbb{E}[e^{-aU}]$ and $L_{\tilde{U}}(a) = \mathbb{E}[e^{-a\tilde{U}}]$ denote the Laplace transforms of U and $\tilde{U}$, respectively. Conditioning on $\mathbf{V} = (V_1, \dots, V_d)$ and using independence yields

$$\mathbb{E}\left[L_U\left(\sum_{j=1}^{d} t_j\, \lambda_j V_j\right)\right] = \mathbb{E}\left[L_{\tilde{U}}\left(\sum_{j=1}^{d} t_j\, \lambda_j \tilde{V}_j\right)\right], \qquad \mathbf{t} \approx \mathbf{0}. \tag{17}$$

We first identify the distribution of the shared frailty. Taking $\mathbf{t} = (t, 0, \dots, 0)$ in (17) gives

$$\mathbb{E}\big[L_U\big(t\, \lambda_1 V_1\big)\big] = \mathbb{E}\big[L_{\tilde{U}}\big(t\, \lambda_1 \tilde{V}_1\big)\big], \qquad t \approx 0,$$

and analogous identities hold for each coordinate. Using also identities obtained for linear combinations of $(V_1, \dots, V_d)$ and exploiting the independence and support of the V_j, we conclude that $L_U = L_{\tilde{U}}$ on an interval containing the origin, and hence everywhere on $\mathbb{R}_+$ by analytic continuation. By uniqueness of Laplace transforms on $(0,\infty)$, this implies $G_0 = \tilde{G}_0$.

With $G_0 = \tilde{G}_0$, identity (17) reduces to

$$\mathbb{E}\left[L_U\left(\sum_{j=1}^{d} t_j\, \lambda_j V_j\right)\right] = \mathbb{E}\left[L_U\left(\sum_{j=1}^{d} t_j\, \lambda_j \tilde{V}_j\right)\right], \qquad \mathbf{t} \approx \mathbf{0}.$$

Since L_U is strictly decreasing and continuous on $\mathbb{R}_+$ whenever $\mathrm{Var}(U) > 0$, this implies equality in distribution of the random sums

$$\sum_{j=1}^{d} t_j \lambda_j V_j \overset{d}{=} \sum_{j=1}^{d} t_j \lambda_j \tilde{V}_j, \qquad \forall \mathbf{t} \approx \mathbf{0}.$$

By the Cramér–Wold device applied on $\mathbb{R}^d$, we obtain

$$(\lambda_1 V_1, \dots, \lambda_d V_d) \overset{d}{=} (\lambda_1 \tilde{V}_1, \dots, \lambda_d \tilde{V}_d).$$

Finally, since the V_j and $\tilde{V}_j$ are independent and $\lambda_j > 0$ are known, this implies $V_j \overset{d}{=} \tilde{V}_j$ for all j, that is, $G_j = \tilde{G}_j$ for $j = 1, \dots, d$.

Combining the above arguments yields $\Theta = \widetilde{\Theta}$ and hence identifiability. □

Appendix C. Proof of tail dependence result

Proof of Proposition 5. Let (Y_j, Y_k) and (Y'_j, Y'_k) be two independent copies of the bivariate subvector of interest. Recall that Kendall's tau can be written as

$$\tau_{jk} = \mathbb{P}\Big\{(Y_j - Y'_j)(Y_k - Y'_k) > 0\Big\} - \mathbb{P}\Big\{(Y_j - Y'_j)(Y_k - Y'_k) < 0\Big\}.$$

Equivalently, defining $\Delta_j = \mathbf{1}\{Y_j > Y'_j\} - \mathbf{1}\{Y_j < Y'_j\}, \qquad \Delta_k = \mathbf{1}\{Y_k > Y'_k\} - \mathbf{1}\{Y_k < Y'_k\},$

we have $\tau_{jk} = \mathbb{E}(\Delta_j \Delta_k)$. Indeed, $\Delta_j \Delta_k = 1$ on the event of concordance, $\Delta_j \Delta_k = -1$ on the event of discordance, and $\Delta_j \Delta_k = 0$ whenever a tie occurs in at least one coordinate.

Let (U, V_j, V_k) and (U', V'_j, V'_k) denote the latent frailties corresponding to (Y_j, Y_k) and (Y'_j, Y'_k), respectively. Since the two observations are independent, the vectors (U, V_j, V_k) and (U', V'_j, V'_k) are independent. Moreover, conditional on $(U, V_j, V_k, U', V'_j, V'_k)$, the random variables Y_j, Y_k, Y'_j, Y'_k are independent, with

$$Y_\ell \mid (U, V_\ell) \sim \mathrm{Poisson}(\lambda_\ell U V_\ell), \quad Y'_\ell \mid (U', V'_\ell) \sim \mathrm{Poisson}(\lambda_\ell U' V'_\ell), \quad \ell \in \{j, k\}.$$

For $\ell \in \{j, k\}$, define $\psi_\ell(u, u') = \mathbb{E}(\Delta_\ell \mid U = u, U' = u')$. By iterated expectation and conditional independence across coordinates given (U, U'), we obtain

$$\tau_{jk} = \mathbb{E}\Big[\mathbb{E}(\Delta_j \Delta_k \mid U, U')\Big] = \mathbb{E}\Big[\mathbb{E}(\Delta_j \mid U, U')\,\mathbb{E}(\Delta_k \mid U, U')\Big] = \mathbb{E}\Big[\psi_j(U, U')\psi_k(U, U')\Big].$$

It therefore suffices to study the sign of $\psi_\ell(u, u')$. Fix $\ell \in \{j, k\}$. Conditional on $U = u$, the distribution of Y_ℓ is a mixed Poisson distribution with random mean $\lambda_\ell u V_\ell$. Since $V_\ell > 0$ almost surely and $\lambda_\ell > 0$, the conditional law of Y_ℓ given $U = u$ is stochastically increasing in u: if $u_1 < u_2$, then

$$Y_\ell \mid U = u_1 \quad \leq_{\mathrm{st}} \quad Y_\ell \mid U = u_2.$$

Consequently, if $u > u'$, then $\mathbb{P}(Y_\ell > Y'_\ell \mid U = u, U' = u') > \mathbb{P}(Y_\ell < Y'_\ell \mid U = u, U' = u')$, so that $\psi_\ell(u, u') > 0$. Similarly, if $u < u'$, then $\psi_\ell(u, u') < 0$, and if $u = u'$, then by symmetry $\psi_\ell(u, u') = 0$. Hence

$$\mathrm{sgn}\big(\psi_\ell(u, u')\big) = \mathrm{sgn}(u - u'), \quad \ell \in \{j, k\}.$$

It follows that for all (u, u'), $\psi_j(u, u')\psi_k(u, u') \geq 0$, with strict inequality whenever $u \neq u'$, because both factors then have the same nonzero sign. Therefore, $\tau_{jk} = \mathbb{E}\Big[\psi_j(U, U')\psi_k(U, U')\Big] \geq 0$. If $\mathrm{Var}(U) = 0$, then U is degenerate almost surely, so $U = U'$ almost surely and thus $\psi_j(U, U') = \psi_k(U, U') = 0$ almost surely. Hence $\tau_{jk} = 0$. If $\mathrm{Var}(U) > 0$, then U is non-degenerate, so $\mathbb{P}(U \neq U') > 0$. On the event $\{U \neq U'\}$, we have $\psi_j(U, U')\psi_k(U, U') > 0$. Therefore, $\tau_{jk} = \mathbb{E}\Big[\psi_j(U, U')\psi_k(U, U')\Big] > 0$. This proves that Kendall's tau vanishes exactly when the shared frailty is degenerate and is strictly positive whenever the shared frailty has positive variance. Hence the shared frailty U is the unique source of pairwise concordance between the count variables. □

Proof of Theorem 3. We prove that the upper tail dependence coefficient

$$\lambda_U = \lim_{t\to\infty} \mathbb{P}(Y_j > t \mid Y_k > t)$$

exists and is strictly positive under the stated assumptions.

First, conditional on (U, V_j, V_k), the variables Y_j and Y_k are independent Poisson random variables with respective means $\lambda_j U V_j$ and $\lambda_k U V_k$. Consequently,

$$\mathbb{P}(Y_j > t, Y_k > t) = \mathbb{E}\left[\mathbb{P}\left(\text{Poisson}(\lambda_j U V_j) > t\right) \mathbb{P}\left(\text{Poisson}(\lambda_k U V_k) > t\right)\right].$$

Next, recall a classical tail equivalence result for Poisson mixtures. Let $Z \sim \text{Poisson}(W)$, where W is a nonnegative random variable with regularly varying tail

$$\mathbb{P}(W > w) \sim w^{-\rho} L(w), \qquad w \to \infty,$$

for some $\rho > 0$ and slowly varying function L. Then (see, e.g., Foss et al., 2013, Chapter 4),

$$\mathbb{P}(Z > t) \sim \mathbb{P}(W > t), \qquad t \to \infty. \tag{18}$$

In the present setting, define $W_j := \lambda_j U V_j$ and $W_k := \lambda_k U V_k$. Since U is regularly varying with index ρ and V_j, V_k have finite moments, Breiman's lemma yields

$$\mathbb{P}(W_j > t) \sim \lambda_j^\rho \mathbb{E}[V_j^\rho]\, \mathbb{P}(U > t), \qquad \mathbb{P}(W_k > t) \sim \lambda_k^\rho \mathbb{E}[V_k^\rho]\, \mathbb{P}(U > t).$$

Combining these relations with (18), we obtain

$$\mathbb{P}(Y_j > t) \sim C_j\, \mathbb{P}(U > t), \qquad \mathbb{P}(Y_k > t) \sim C_k\, \mathbb{P}(U > t),$$

for strictly positive constants C_j and C_k.

We now turn to joint exceedances. Using conditional independence and the same tail equivalence arguments,

$$\mathbb{P}(Y_j > t, Y_k > t) = \mathbb{E}\left[\mathbb{P}(\text{Poisson}(W_j) > t)\mathbb{P}(\text{Poisson}(W_k) > t)\right].$$

For large t, simultaneous exceedances occur primarily when the shared frailty U takes large values. Applying Breiman's lemma again yields

$$\mathbb{P}(Y_j > t, Y_k > t) \sim C_{jk}\, \mathbb{P}(U > t),$$

for some finite, strictly positive constant C_{jk}.

Finally, combining the marginal and joint asymptotics gives

$$\lambda_U = \lim_{t\to\infty} \frac{\mathbb{P}(Y_j > t, Y_k > t)}{\mathbb{P}(Y_k > t)} = \frac{C_{jk}}{C_k} > 0,$$

which completes the proof. □

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